Chapter 5 Water Quality
CIVL1105 Environmental Engineering



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The University of Hong Kong (HKU



Topics and Schedule

Week	Tuesday (9:00-9:50) MB237	Friday (13:00 – 14:50) MB237
1	Introduction to Environmental Engineering	Mass Balance
2	Tutorial (Yabo)	Energy Fundamentals (Dr. Peng)
3	Environmental Chemistry (Dr. Peng)	Environmental Chemistry (Dr. Peng)
4	Tutorial (Zhiheng + Bin)	Water Pollution
5	Water Pollution	Drinking Water Treatment
6	Drinking Water Treatment	Wastewater Treatment Overview and Preliminary/primary Treatment
7	Reading week	
8	Biological Wastewater Treatment Principals	Cancelled*
9	Biological Wastewater Treatment Principals	Activated Sludge Process
10	Activated Sludge Process & BNR	Biological nutrient removal
11	Tutorial (Yuantian)	Solid waste management and resource recovery
12	Solid waste management and resource recovery	Air Pollution
13	Air Pollution	Climate Change & Course final review

*As approved by the Senate, **all UG and TPG classes scheduled on October 23, 2026 [Fri] afternoon (starting from 1:00 pm onwards) and October 24, 2026 [Sat] will be CANCELLED to facilitate the necessary preparations for the Information Day events**. Hence, no classes will be scheduled.

Topics and Schedule

Teaching Weeks 2026-27 for Undergraduate and Taught Postgraduate Students

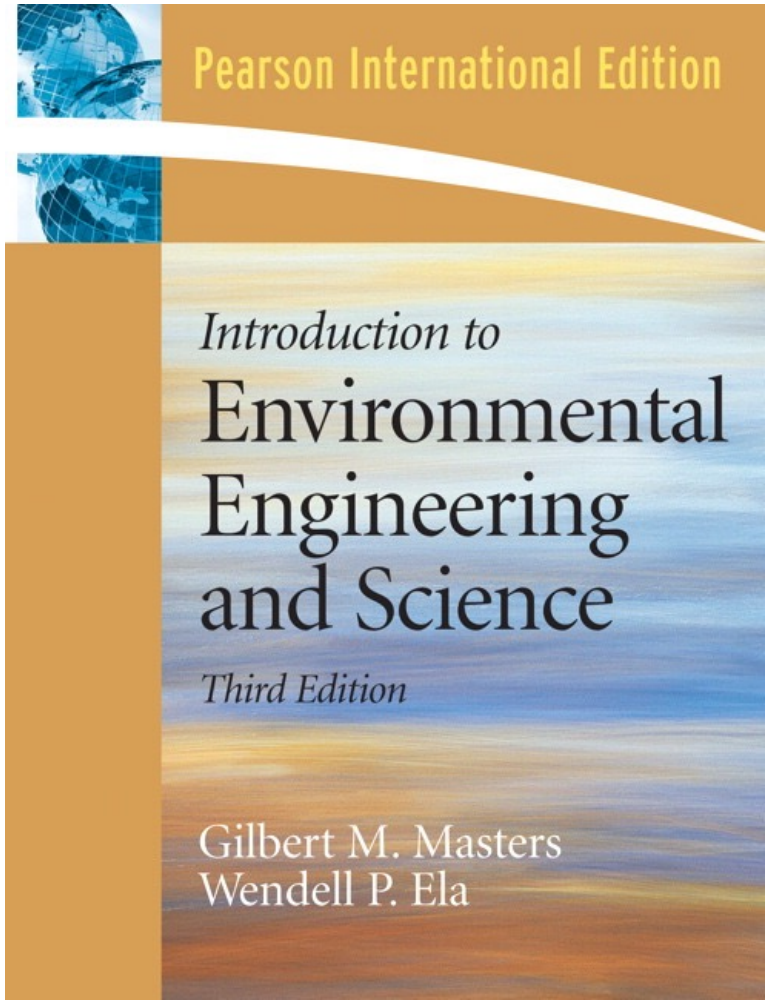
	SUN	MON	TUE	WED	THUR	FRI	SAT	FIRST SEMESTER: SEP 1 - DEC 23, 2026	Week
SEP-26			1	2	3	4	5	First Day of Teaching: Sep 1, 2026	1
	6	7	8	9	10	11	12		2
	13	14	15	16	17	18	19		3
	20	21	22	23	24	25	[26]		4
	27	28	29	30					5
OCT-26					[1]	2	3		
	4	5	6	7	8	9	10		6
	11	12	13	14	15	16	17	Reading/Field Trip Week: Oct 12 - 17, 2026	7(Reading)
	18	[19]	20	21	22	23	24		8
	25	26	27	28	29	30	31		9
NOV-26	1	2	3	4	5	6	7		10
	8	9	10	11	12	13	14		11
	15	16	17	18	19	20	21		12
	22	23	24	25	26	27	28		13
	29	30						Last Day of Teaching: Nov 30, 2026	
DEC-26			1	2	3	4	5	Revision Period: Dec 1 - 5, 2026	14(Revision)
	6	7	8	9	10	11	12	Assessment Period: Dec 7 - 23, 2026	1
	13	14	15	16	17	18	19		2
	20	21	22	23	(24)	[25]	[26]		3
	27	28	29	30	<31>				Break

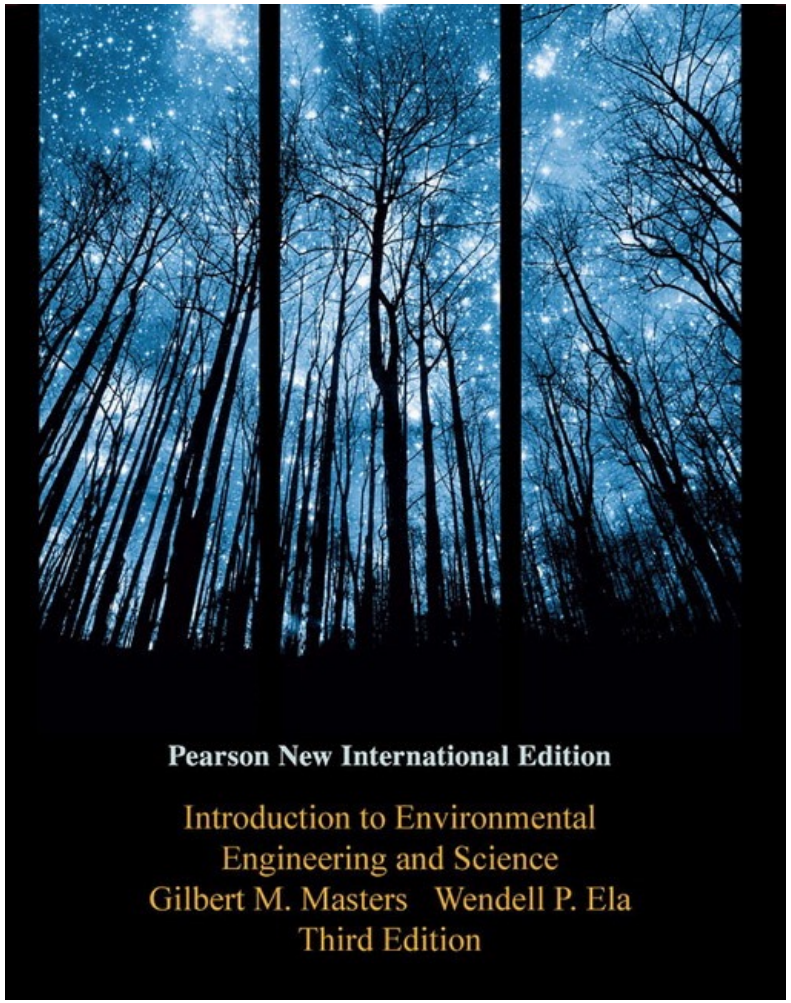
Note:
No class scheduled in reading week.

No class on 23rd Oct as per university arrangement

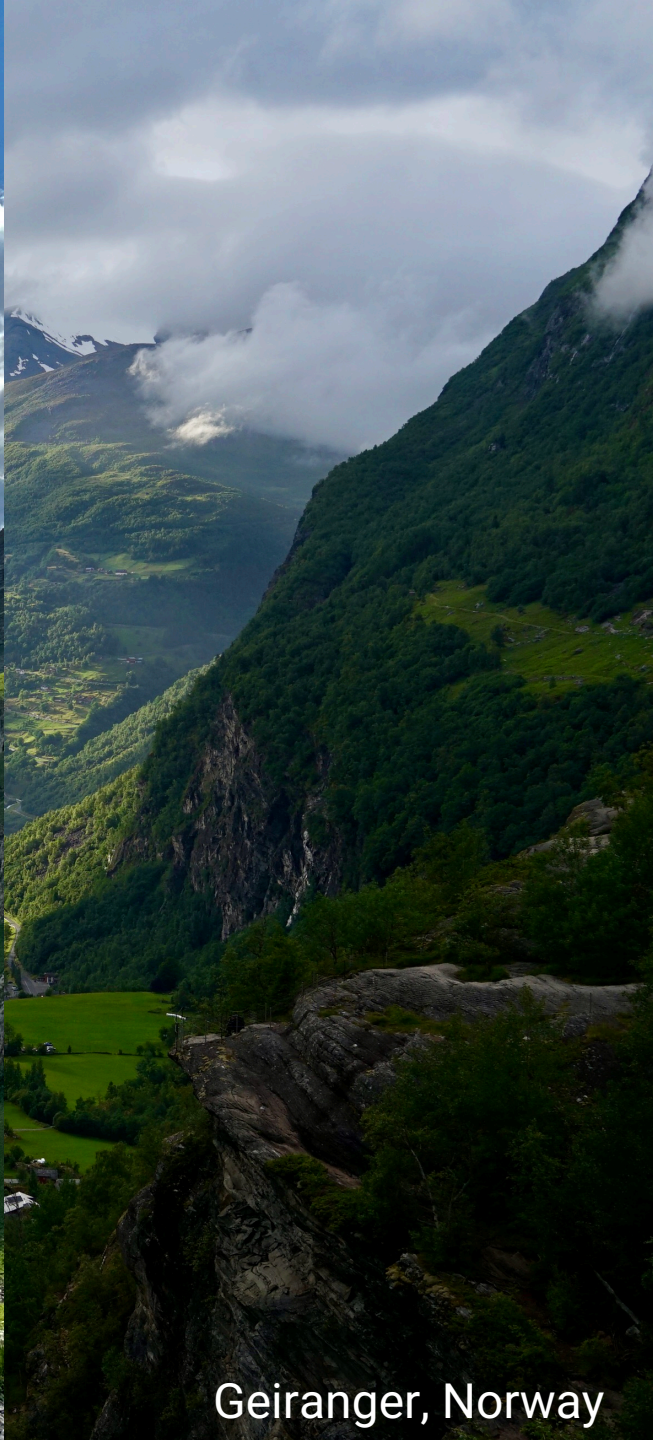
Textbook

Introduction to Environmental Engineering and Science: International Edition (3e) by Gilbert Masters and Wendell Ela






Oeschinensee, Switzerland


Geiranger, Norway





Expected learning outcomes for Chapter 5

- Properties of water
- Water pollutants (physical, chemical, and biological)
- **DO and BOD**
  - **Oxygen solubility, Henry's Law**
  - **BOD5, ultimate BOD, BOD modeling, and temperature correction**
  - CBOD vs. NBOD
  - BOD vs. other parameters for indicating organic waste strength
- DO in rivers
  - Oxygen sag curve
- Eutrophication
  - Eutrophication, nutrients (N & P), limiting nutrient
- Thermal stratification

Why water quality matters?

WATER QUALITY MANAGEMENT

- **prevention of pollution** in surface and groundwater
- ensuring adequate **water availability** to users
- optimization of **water quality** for all beneficial users

It involves

- knowledge of effects of pollutants on water bodies
- ability to assess responses of water bodies to waste load
- knowledge and application of measures to abate water pollution

mad max fury road

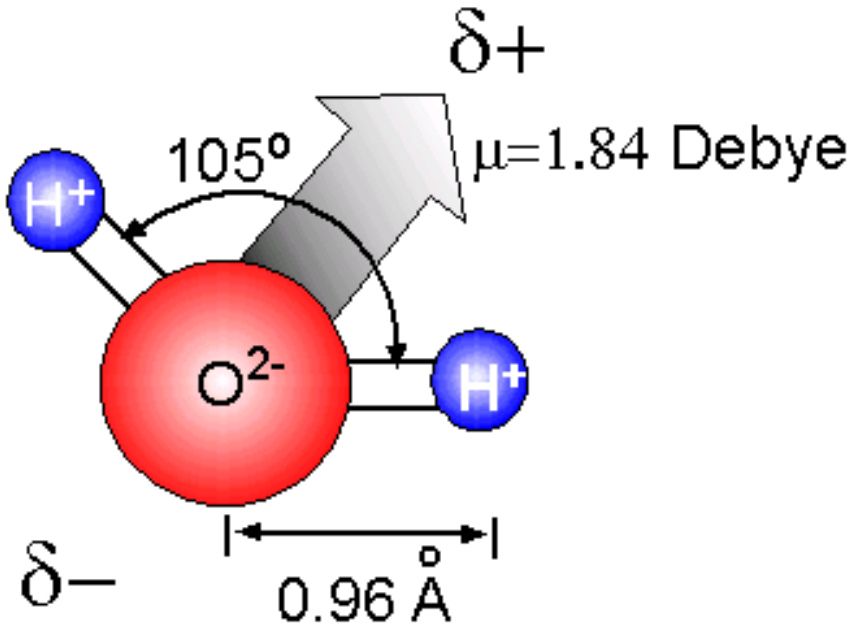
5.1 Water and water pollutants

- ❑ Properties of water
- ❑ Water pollutants

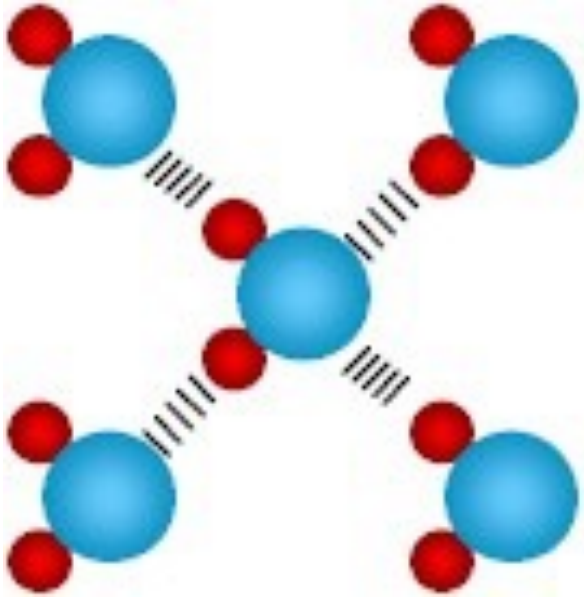
Water as a Solvent

Micro-Level

Structure of Water: Dipole



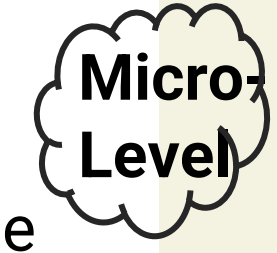
Hydrogen bonding



http://www.visionlearning.com/en/library/Chemistry/1/Water/57

Further reading:
http://www.visionlearning.com/en/library/Chemistry/1/Water/57
http://www.chem1.com/acad/sci/aboutwater.html

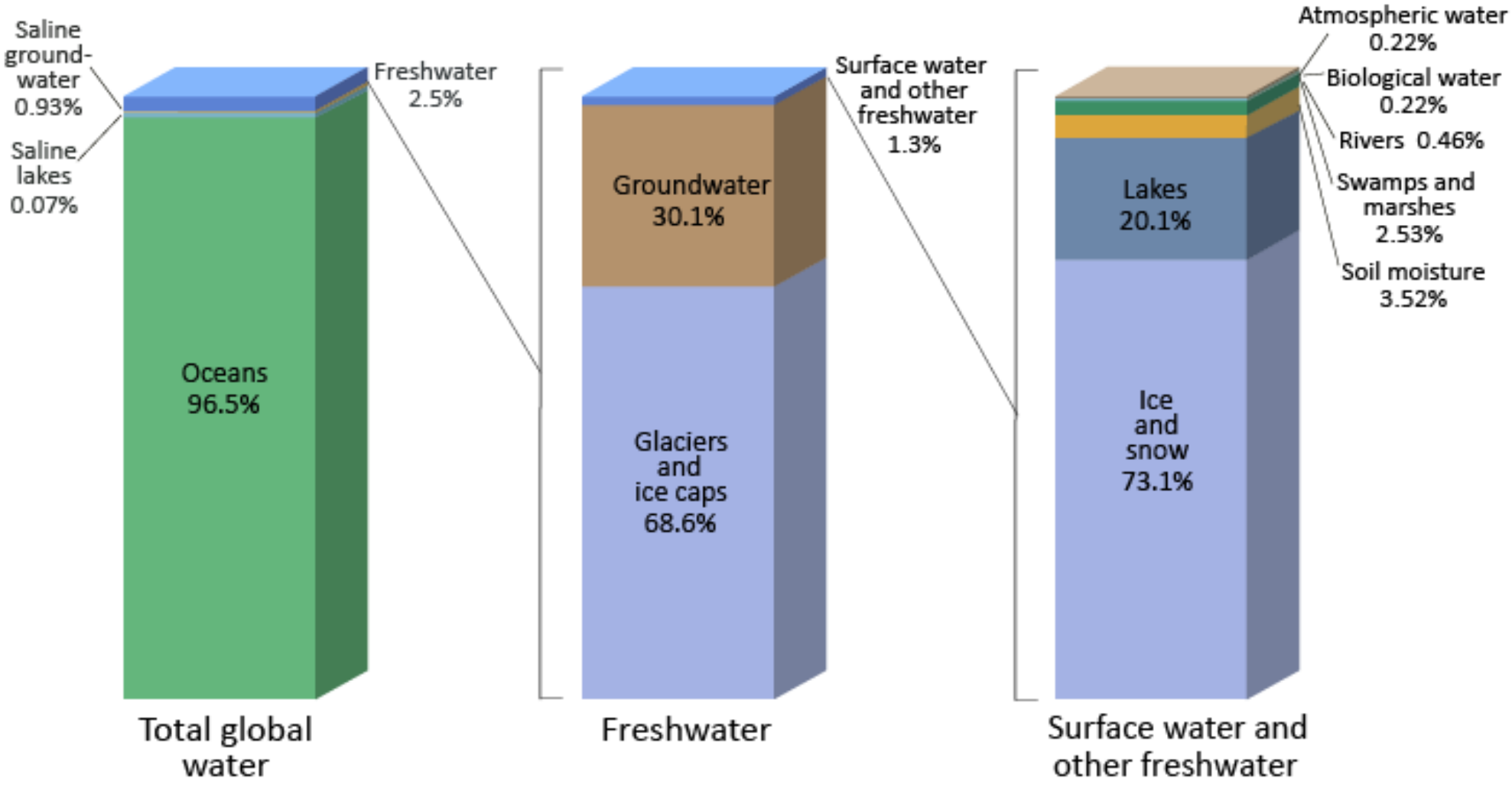
Dipoles



- **Dipoles** occur when electrons are shared unequally between atoms in the same molecule due to a high difference in the electronegativity of the atoms involved.
- A dipole is a molecule or covalent bond that has a separation of charges.

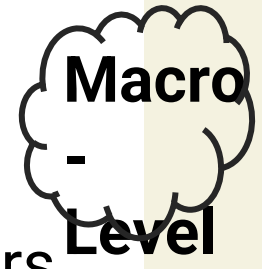
- **Hydrogen bonding** is a special type of dipole-dipole attraction between molecules, not a covalent bond to a hydrogen atom. It results from the attractive force between a hydrogen atom covalently bonded to a very electronegative atom such as a N, O, or F atom and another very electronegative atom.

Distribution of Earth's Water



Source: Igor Shiklomanov's chapter "World fresh water resources" in Peter H. Gleick (editor), 1993, Water in Crisis: A Guide to the World's Fresh Water Resources.

Macro - Level

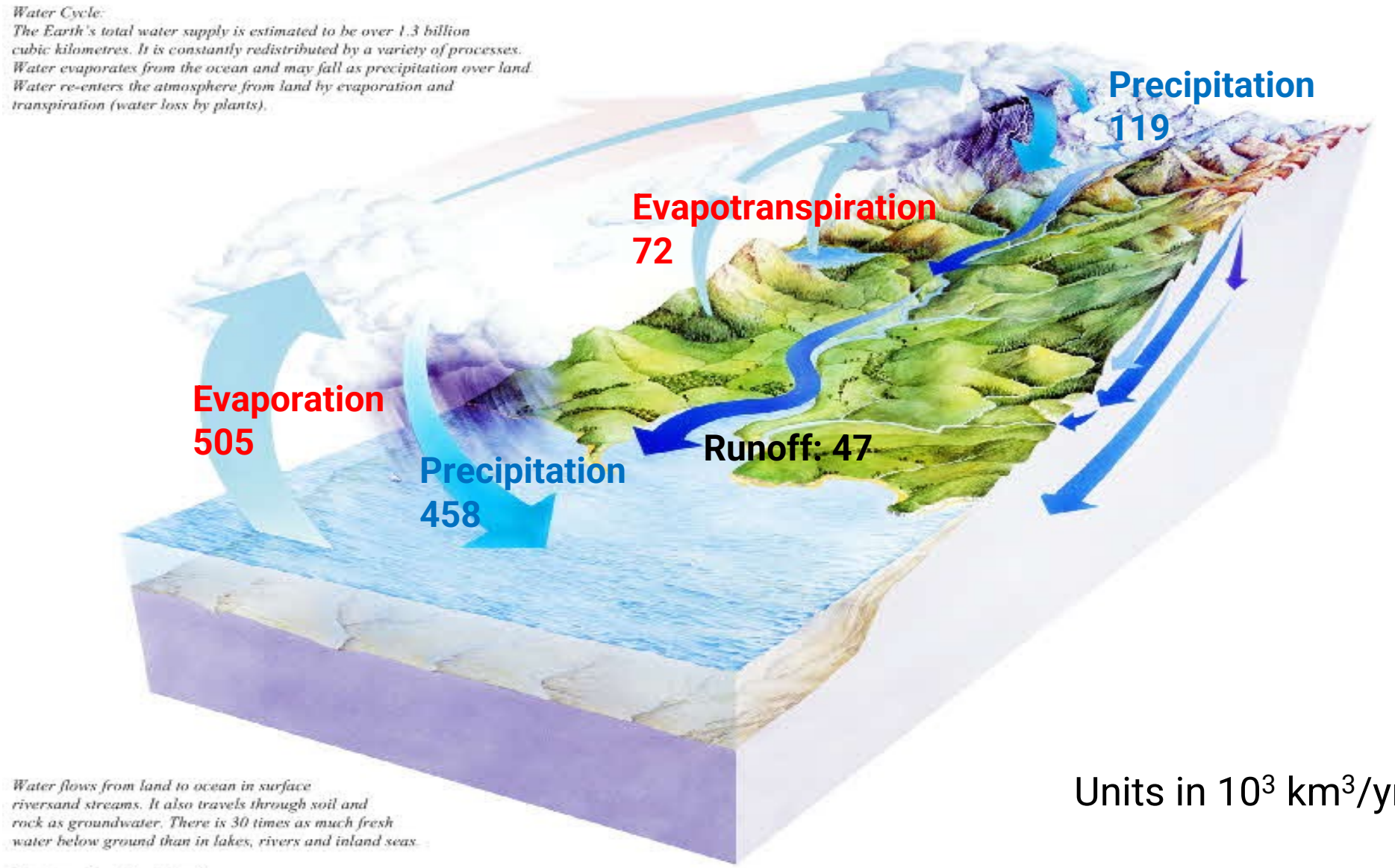


- **Surface water** is any body of water above ground, including streams, rivers, lakes, wetlands, reservoirs, and creeks. The ocean, despite being saltwater, is also considered surface water.

- **Groundwater** is fresh water located in the subsurface pore space of soil and rocks. It is also water that is flowing within aquifers below the water table. Sometimes it is useful to make a distinction between groundwater that is closely associated with surface water, and deep groundwater in an aquifer. About 30 percent of all readily available freshwater in the world is groundwater

The hydrologic cycle

Macro - Level



Units in 10^3 km^3/yr

Source: USGS

Water usage

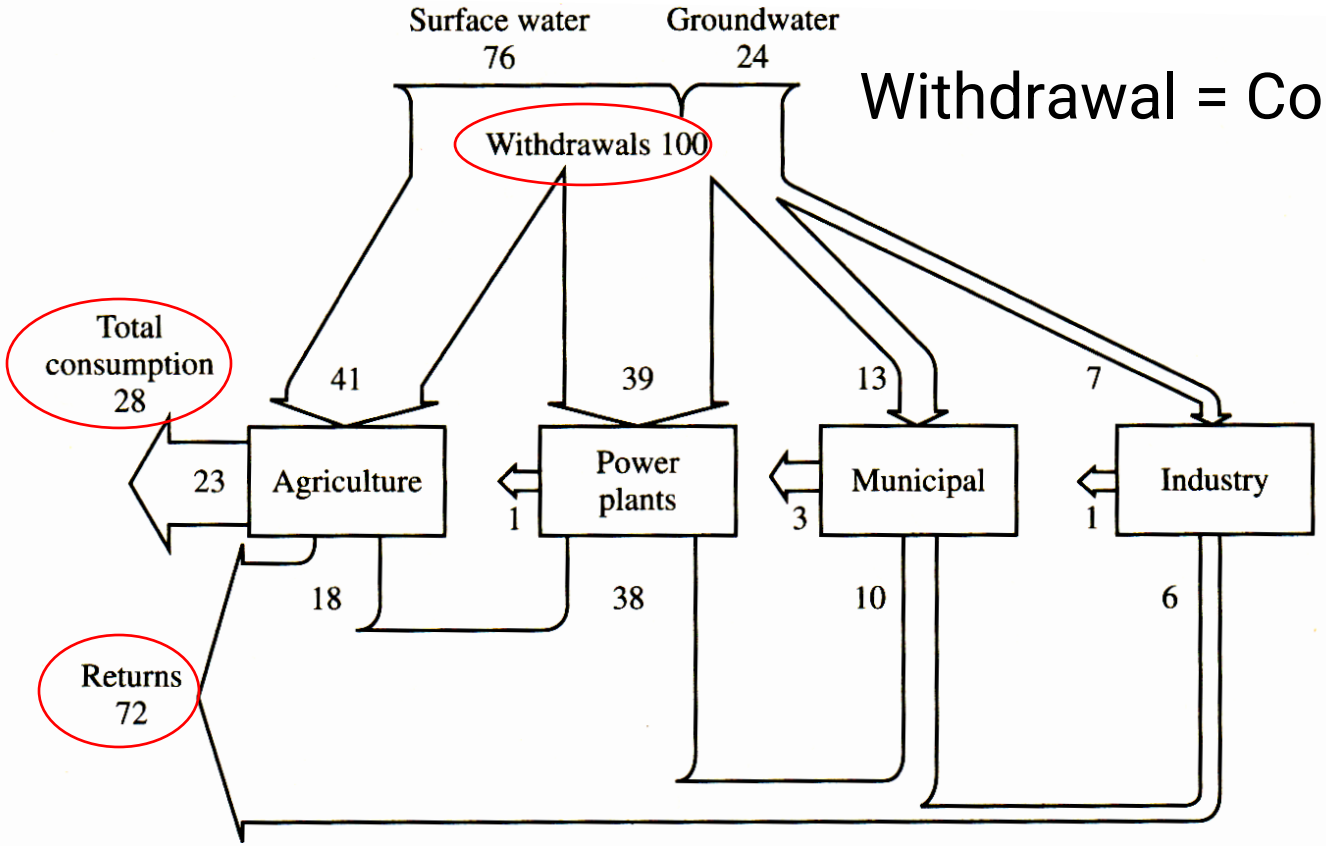
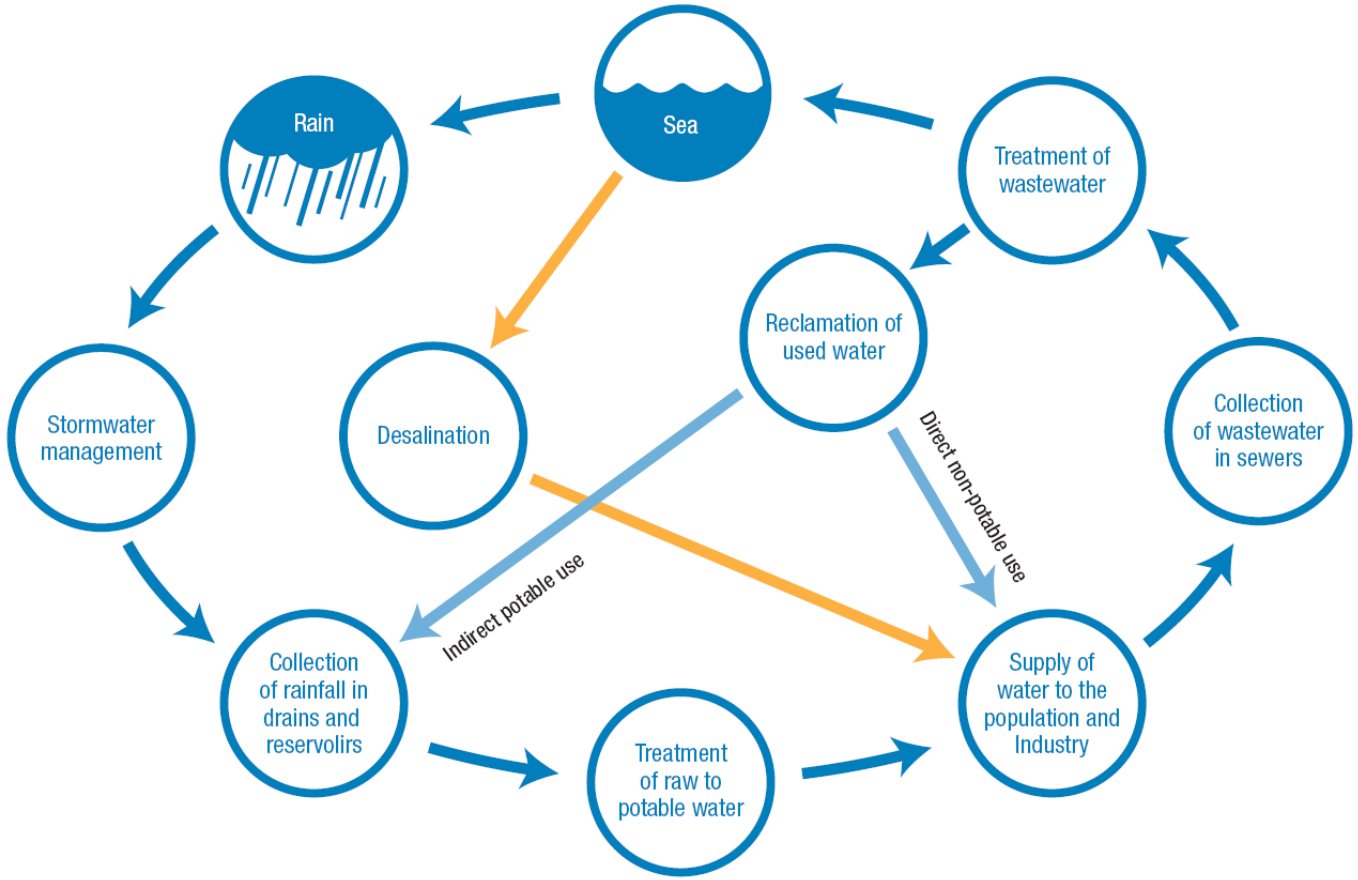


FIGURE 5.5 Fresh water use in the United States, 1990. The proportion of withdrawals by category of use was unchanged in 2000. Annual 2000 freshwater withdrawals are about 476 km^3.
(*Sources:* From Gleick, 1993; U.S. Water Resources Council, 1978; and Hutson et al., 2004.)
Notes: Industry includes mining, *agriculture* includes irrigation and livestock, and *municipal* includes public, domestic, and commercial uses.

Withdrawal = Consumption + Returns

Textbook page 180

Water cycle



Water
Source of life!



Innovation in Water, Issue 1,
http://www.pub.gov.sg/mpublications/Innovation/Documents/June2011.pdf

Water pollutants





Water pollution?





Water pollutants









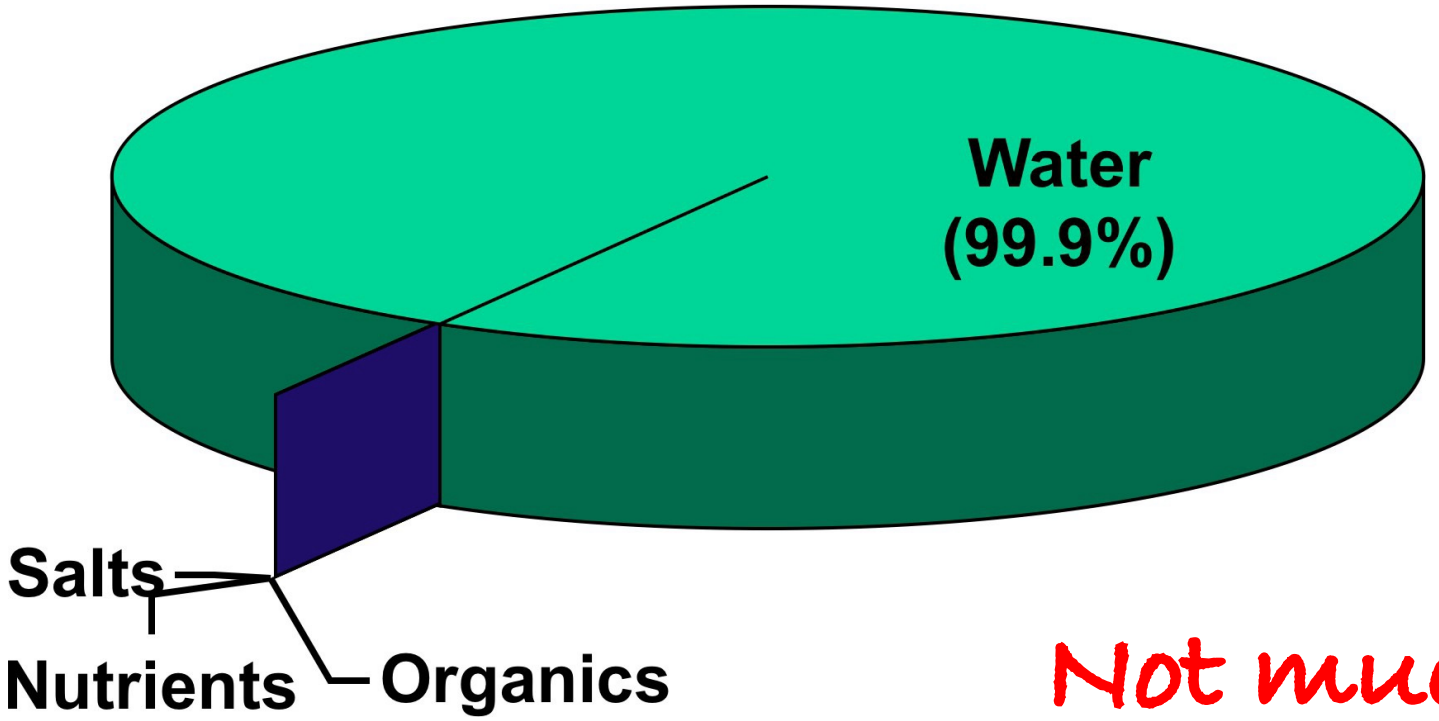
Water pollution?





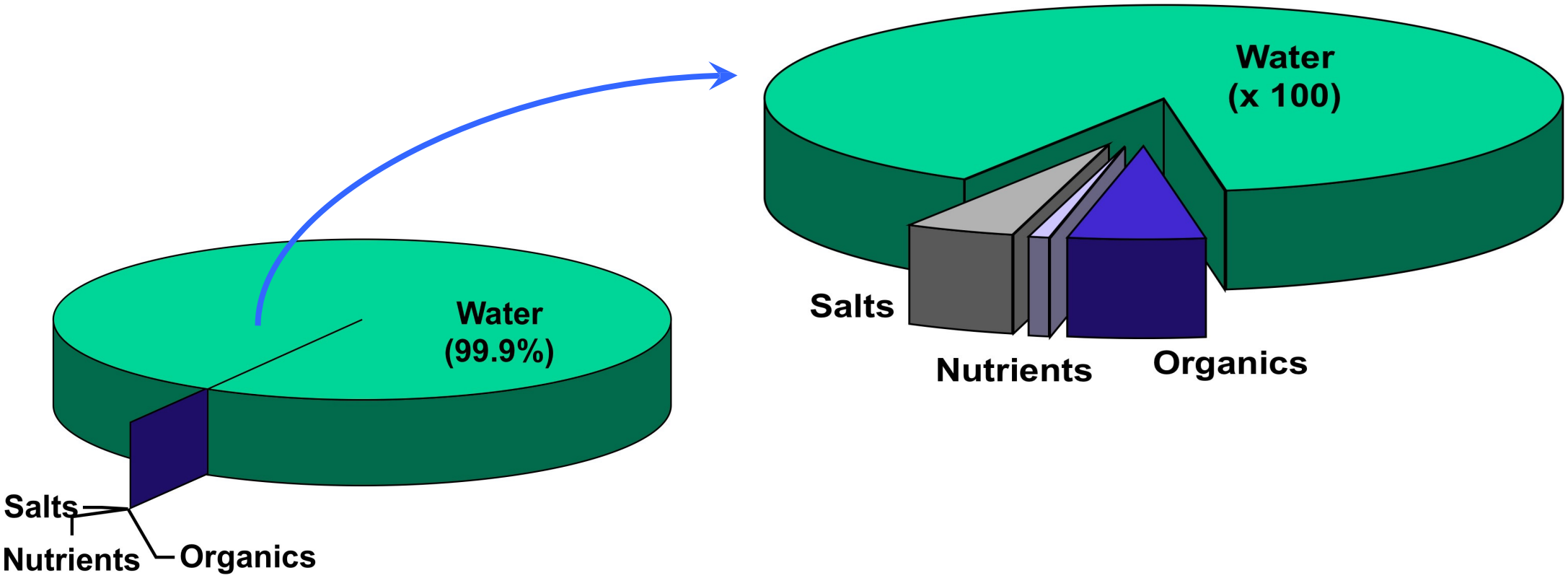


How much “water” is in wastewater?



Not much contaminants !!

What is in wastewater?



Water pollutants

Physical

- Turbidity
- Color
- Taste and Odor

Aesthetic effect

- Temperature → effect on aquatic life (e.g., cooling tower)

Chemical

- **Oxygen demanding wastes** → DO depletion
- **Nutrients** → algal bloom
- Heavy metals → toxic effects
- Pesticides, volatile organic chemicals, emerging contaminants → toxic effects

Biological

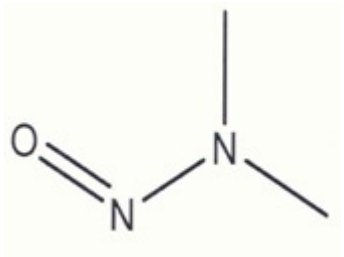
- Pathogen: Disease causing organisms
- Virus, bacteria, protozoa, helminthes

Micropollutants

Organic toxins

- Pharmaceuticals
- Endocrine disruptors
- Pesticides/herbcides
- Disinfection byproducts
- Trace metals
- Other emergent contaminants

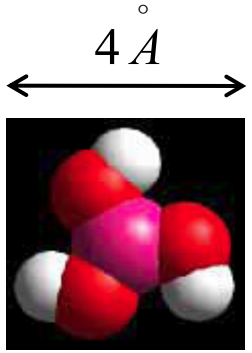




N-nitrosodimethylamine (NDMA)

Inorganics

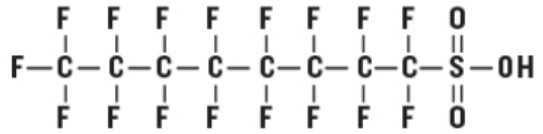
- Boron
- Bromide
- …



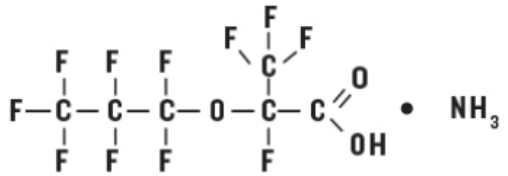
Boric acid



PFOA



PFOS



GenX

Heavy metals ➡ trace metals

NEWS • HONG KONG • HEALTH & ENVIRONMENT • HEALTH

Give water filters to Hong Kong families hit by lead contamination scare, government urged

Gloria Chan
gloria.chan@scmp.com

PUBLISHED : Friday, 24 July, 2015, 12:01pm
UPDATED : Friday, 24 July, 2015, 10:52pm



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2015年7月11日 星期六 下一篇

啟晴食水鉛超標最高逾倍 房署認驗樓沒驗水 政黨促查承建商其他工程

723



Water checks widen as more lead found

Amy Nip

The lead-in-water scare widened yesterday, with Wing Cheong Estate in Sham Shui Po becoming the latest victim.

The discovery comes as the Secretary for Transport and Housing Anthony Cheung Bing-leung announced that water checks are to be extended to another 12 public housing estates that were completed in 2011 and 2012.

"We have decided to expand the scope to all estates completed since 2011," Cheung said. "It involves an additional 12 estates comprising 35 blocks and 26,000 flats."

After lead in water was found at Kai Ching Estate in Kowloon City, Kwai Luen in Kwai Tsing and Shui Chuen O in Sha Tin, the government collected water from another nine estates, including Wing Cheong and phase one of Kwai Luen Estate, for checks last week.



Wing Cheong is the latest public estate dragged into the contaminated water scandal. SING TAO

water, exceeding the World Health Organization limit of 10 micrograms.

contractor of Wing Cheong Estate and Golden Day Engineering was responsible for pipe works. The plumber was not named yesterday.

The companies are different to those responsible for the construction at Kai Ching and Kwai Luen. At Wing Cheong, water tanks have been arranged for each block and bottled water is provided for the vulnerable.

The 12 estates to have their water tested are Shin Ming, Tin Ching, Sha Tin Pass, Yan On, Choi Tak, Choi Fook, Yau Lai, Hung Hom, Tung Wui, Shek Kip Mei Estate, Lower Ngau Tau Kok and Un Chau.

Cheung said after tests at the additional 12 estates were completed, he would not rule out further extending the coverage of water tests to more public estates.

Secretary for Food and Health Ko Wing-man said 924 people have been booked for blood tests. Some may have to wait until next month.

*** WHO guideline values 世界衛生組織建議標準**

	µg/L 每公升含量(微克)
Lead 鉛	10
Cadmium 鎘	3
Chromium 鉻	50
Nickel 鎳	70

Example: Lead

Source

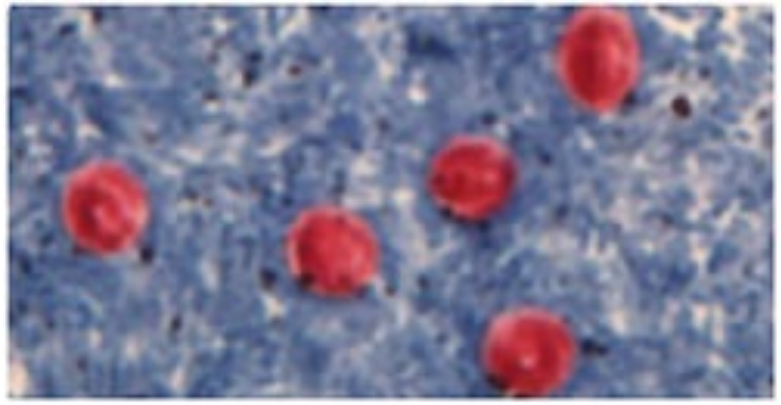
- Mainly from leaded gasoline
- Emitted mainly in the form of inorganic particulates, settling down in vicinity of source (e.g., highways)
- Switch to non-leaded gasoline → greatly reduced lead emission
- Other sources: leaded paint, pipelines (leaded solder at joints)
- Exposure
  - Inhalation
  - Water
- Health effect
  - Reduced intelligence and detrimental effects on growth and development of children
  - Brain damage

Pathogen

1993 Milwaukee Cryptosporidiosis outbreak

- Caused by Cryptosporidium
- the largest waterborne disease outbreak in USA
- 403,000 residents became ill
- 104 deaths

(Source of info: http://en.wikipedia.org/wiki/1993_Milwaukee_Cryptosporidiosis_outbreak)





SATURDAY MORNING April 10, 1993

WATER CRISIS: DAY 3

Chemist, city fight over early signs of impurities

'Boy, is that water raunchy'

Water was cloudy in March

Discolored water with illness-causing microbes likely seeped through the Howard Avenue water purification plant in late March, city officials said.

6A EPA action

Plants to be told to monitor parasite

Marks said, "The warning signs were not recognized or responded to."

"The plant could have been shut down when the turbidity

Pathogen

Outbreak of Giardiasis and Cryptosporidiosis Associated with a Neighborhood Interactive Water Fountain–Florida, 2006

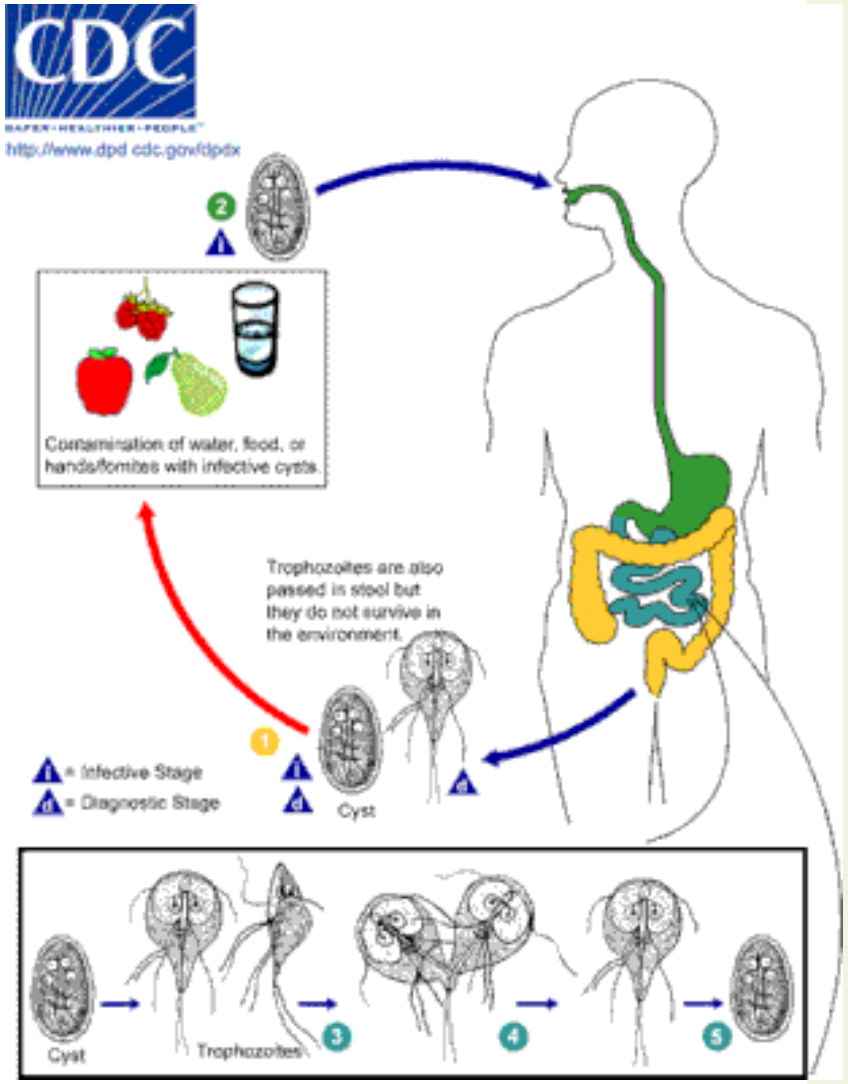
Leah Eisenstein, M.P.H.
Dean Bodager, M.P.A., R.S., D.A.A.S.
Dawn Ginzl, M.P.H.

Abstract

An outbreak of giardiasis and cryptosporidiosis was identified in central Florida in September 2006. Environmental and epidemiological investigations indicated the likely source was a neighborhood interactive water fountain in a large upscale urban neighborhood. Forty-nine cases meeting the case definition were identified, of which 38 were giardiasis, nine were cryptosporidiosis, and two were co-infections. The median age of those affected was four years old, and 32 (65.3%) cases were male. This outbreak and other similar occurrences highlight the need to design and implement more stringent disinfection practices and filtration requirements for treated interactive water venues. *Giardia* cysts and *Cryptosporidium* oocysts are small and chlorine-resistant, and they may require supplemental disinfection methods, such as ultraviolet light irradiation, ozonation, or chlorine dioxide. Individuals who use these types of venues also need to change their behavior to prevent disease transmission. This is the first documentation of a giardiasis outbreak associated with exposure to an interactive water fountain in the United States.

fountains typically consist of multiple jets of water, with little or no standing water, and human interaction is encouraged. These features are increasingly seen in areas surrounding pools and in nontraditional areas, such as outdoor malls, amusement parks, and other community locations. These features are ideal for young children, especially those who are diaper-aged, as there is negligible drowning risk and minimal supervision is required. Heavy use by diaper-aged children increases the risk of fecal contamination (Castor & Beach, 2004). The misperception that these venues use potable fresh water instead of recycled chlorinated nonpotable water may encourage children to intentionally swallow water. The spray from interactive water fountains also increases the opportunities to swallow water (Castor & Beach, 2004).

Six waterborne disease outbreaks resulting

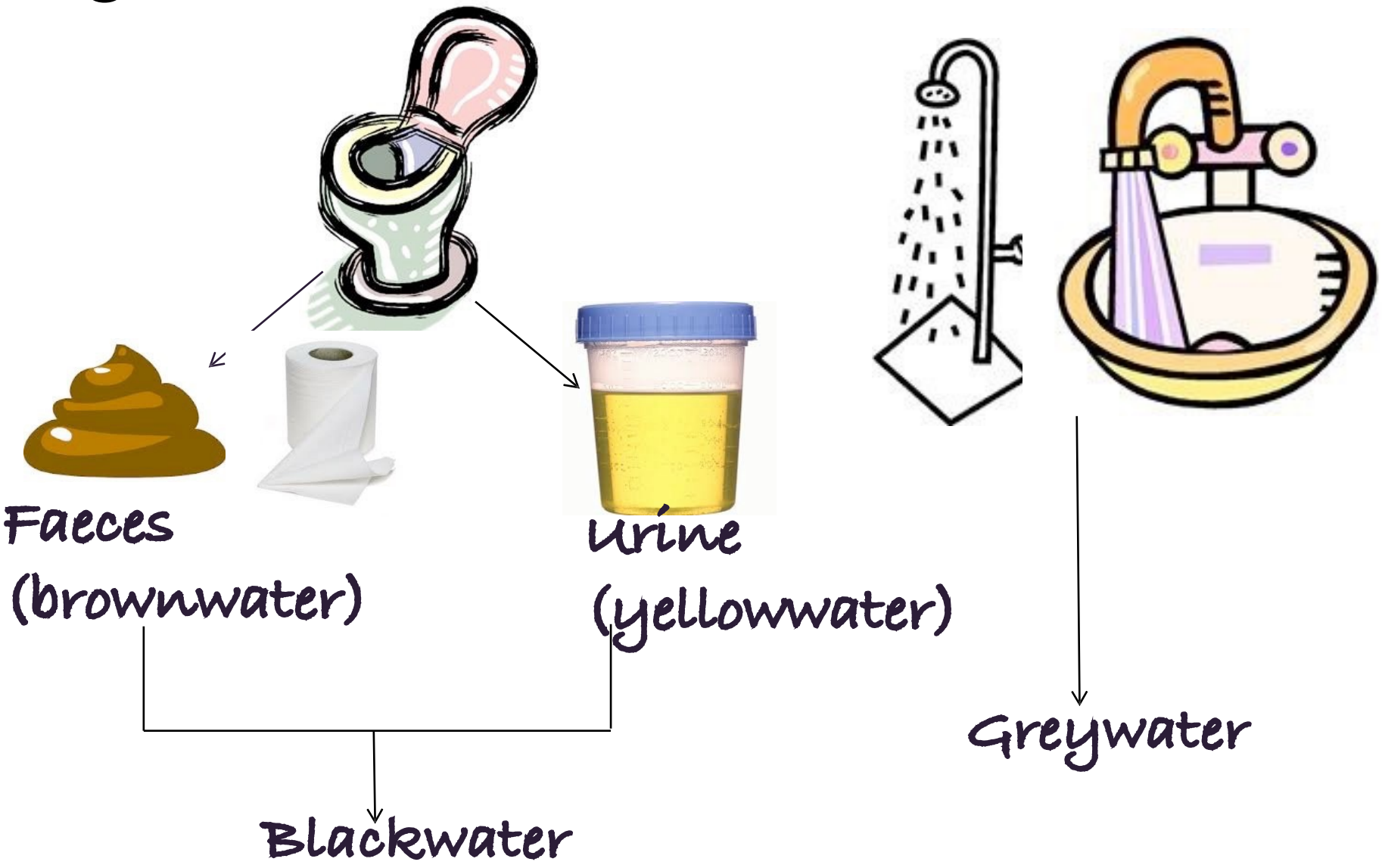


Life cycle of Giardia (Source: CDC)

Major contaminants in domestic wastewater

Constituent	Average concentration	Comments
BOD_5	200 mg BOD/L	O_2 demanding material (carbohydrates, proteins, other organic matter)
Suspended solids	240 mg/L	Cause turbidity, may contribute to BOD
Pathogens	3,000,000 coliforms/100mL	Cause diseases, normally associated with fecal matter
Nutrients (mainly N and P)	Total N: 35 mg N/L Inorganic N: 15 mg/L Total P: 10 mg P/L	Contribute to eutrophication
Toxic chemicals	variable	Heavy metals (Hg, Cd, Cr), pesticides, solvents, etc.
Emerging chemicals of concern	variable	Pharmaceuticals, caffeine, endocrine disruptors, etc.

What goes into sewer from our homes



Some interesting stats about urine…

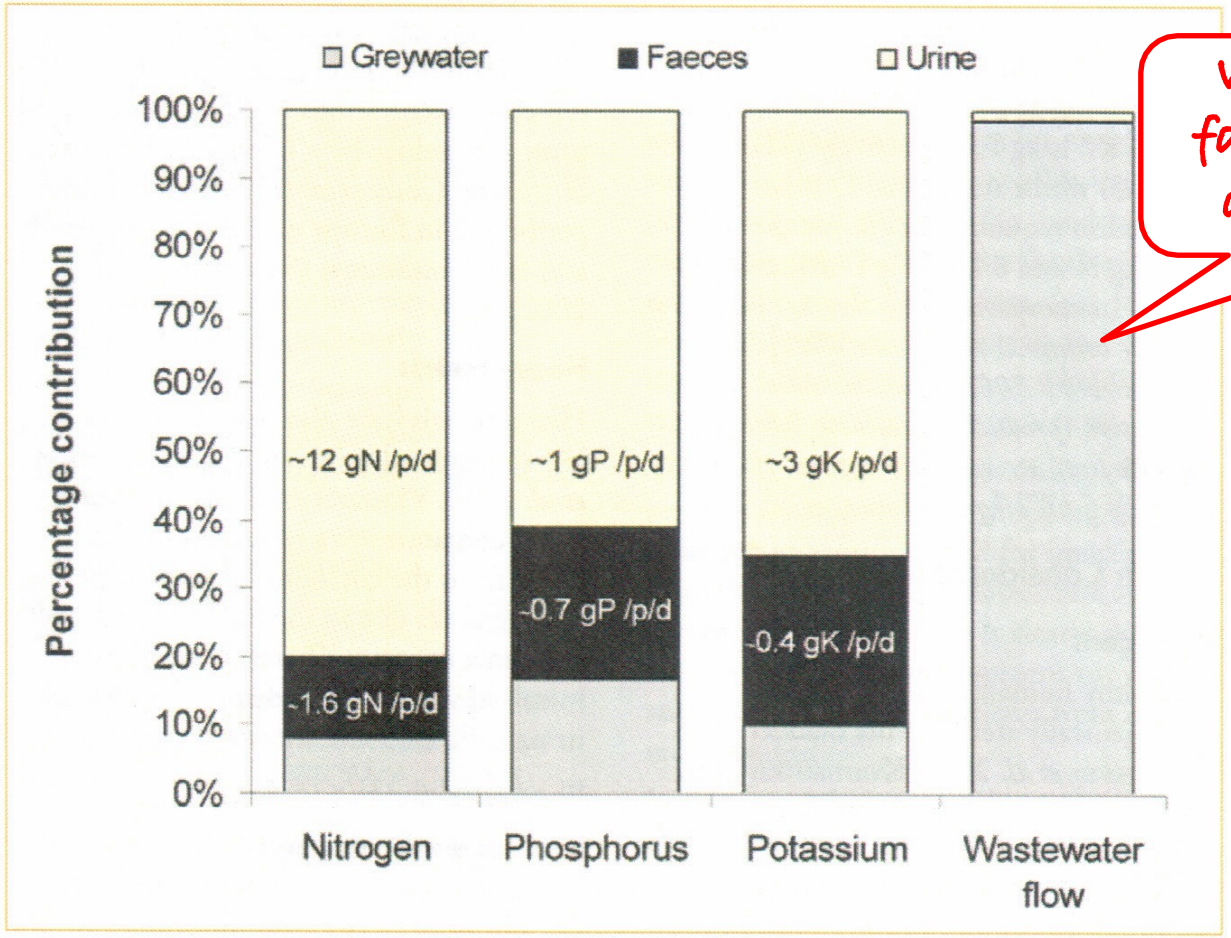


Figure 1. The percentage of nutrients and their daily excreted mass in various wastewater components (Johansson *et al.* 2003).

Why mix it with faeces and flush it down the sewer ?



Urine-diverting toilet (Dena Fam, UTS)
theconversation.com

Taken from Beal, Gardner, Ahmed, Walton and Hamlyn-Harris (2008). 'Urine-Separation and Reuse Trial', WATER, February 2008.

Typical yellowwater loads

Table 17.3 Compounds in urine and faeces.

		Urine			Faeces		
		Av.	Std.	Range	Av.	Std.	Range
Wet mass	[g·p^{-1}·d^{-1}]	1,270	363	1,010–1,530	200	100	100–350
Dry mass	[g·p^{-1}·d^{-1}]	58			41	10	31–53
Water cont.	[%]	95	1.4	94–96	77	7	65–85
pH	[–]	6.4	1.1	5.0–7.2			
EC	[mS·cm^{-1}]	19	2.4	8.7–31			
TSS	[g·p^{-1}·d^{-1}]				22	23	6–60
$BOD_{5\ t}$	[g·p^{-1}·d^{-1}]	5.8	0.7	1.8–10	12	11	4.3–20
COD_t	[g·p^{-1}·d^{-1}]	13	2.6	5.0–24	31	41	2.6–63
TOC	[g·p^{-1}·d^{-1}]	4.3			3.5		
TAN	[mg·p^{-1}·d^{-1}]	550	36	320–880	301	408	13–590
N_{tot}	[g·p^{-1}·d^{-1}]	11	0.3	4–16	1.5	0.0	0.3–4.2
Urea	[g·p^{-1}·d^{-1}]	21	3				
P_t *	[g·p^{-1}·d^{-1}]	0.93		0.8–2.0	0.6		0.3–0.8
K	[g·p^{-1}·d^{-1}]	2.6	0.1	1.0–4.9	0.9	0.2	0.2–1.3
S	[g·p^{-1}·d^{-1}]	1.3		1.2–1.5	0.61	0.19	
Pb	[µg·p^{-1}·d^{-1}]	2.0			20		
Cu	[µg·p^{-1}·d^{-1}]	103			1,110		
Hg	[µg·p^{-1}·d^{-1}]	1.9	1.6	0.8–3.1	8.3		
Zn	[µg·p^{-1}·d^{-1}]	46			11		
FC	[cell·p^{-1}·d^{-1}]						10^8–10^{11}
FS	[cell·p^{-1}·d^{-1}]						10^7–10^{10}
Cps	[cell·p^{-1}·d^{-1}]						10^5–10^{12}

Compilation of data from: Siegrist *et al.* (1976), Ciba-Geiy (1977), Feachem *et al.* (1983), Fittschen and Hahn (1998), Almeida *et al.* (1999), Udert (2002), Udert *et al.* (2003), Vinnerås *et al.* (2006), Etter *et al.* (2011), Meinzinger and Oldenburg (2009). EC – Electrical conductivity; $_t$ – total; TAN – Total ammonium nitrogen; FC – Fecal coliforms; FS – Fecal Streptococci; Cps – *Clostridium perfringens* spores. * About 95% of P_t in urine is phosphate.

Typical brownwater loads
(toilet paper contribution not included)

Taken from: Larsen, Udert and Lienert. Wastewater composition, Chapter 17. In: “Source separation and decentralisation for wastewater management”. IWA Publishing, London, 2013.

Typical greywater loads

Table 17.5 Pollutants in greywater.

	Loads				Concentrations			
		Av.	Std.	Range		Av./Gm.	Std.	Range
pH						7.2	0.52	6.4–10.0
EC					[$\mu S \cdot cm^{-1}$]	1.8	0.66	0.1–2.8
Turb					[NTU]	95	90	20–280
TSS	[$g \cdot p^{-1} \cdot d^{-1}$]	19	21	2–125	[$mg \cdot L^{-1}$]	216	270	2.0–1,070
VSS	[$g \cdot p^{-1} \cdot d^{-1}$]	6.9	8.6	1.6–20	[$mg \cdot L^{-1}$]	64		
COD_t	[$g \cdot p^{-1} \cdot d^{-1}$]	51	24	7.0–102	[$mg \cdot L^{-1}$]	620	707	7–2,570
TOC	[$g \cdot p^{-1} \cdot d^{-1}$]	8.4	6.4	1.8–16	[$mg \cdot L^{-1}$]	83	14	73–93
$BOD_{5\ t}$	[$g \cdot p^{-1} \cdot d^{-1}$]	19	19	1.0–63	[$mg \cdot L^{-1}$]	279	314	1–1,060
$BOD_{5\ d}$	[$g \cdot p^{-1} \cdot d^{-1}$]	7.6	7.7	1.9–19	[$mg \cdot L^{-1}$]	34	3.4	31–36
N_t	[$g \cdot p^{-1} \cdot d^{-1}$]	0.9	0.7	0.007–2.3	[$mg \cdot L^{-1}$]	23	38	0.1–128
TAN	[$g \cdot p^{-1} \cdot d^{-1}$]	0.32	0.49	0.11–1.3	[$mg \cdot L^{-1}$]	28	41	1–75
NO_3-N	[$g \cdot p^{-1} \cdot d^{-1}$]	0.64	1.3	0.007–2.5	[$mg \cdot L^{-1}$]	17		0.1–17
PO_4-P	[$g \cdot p^{-1} \cdot d^{-1}$]	0.14	0.19	0.017–0.41	[$mg \cdot L^{-1}$]	0.4		
P_t	[$g \cdot p^{-1} \cdot d^{-1}$]	0.5	0.4	0–2.2	[$mg \cdot L^{-1}$]	8.5	7.2	0.1–42
Cl^-	[$g \cdot p^{-1} \cdot d^{-1}$]	17	18	4.1–37	[$mg \cdot L^{-1}$]	181	37	9–227
K	[$g \cdot p^{-1} \cdot d^{-1}$]	0.3	0.6	0–4.1	[$mg \cdot L^{-1}$]	10	8.4	0.2–24
S	[$g \cdot p^{-1} \cdot d^{-1}$]	2.9		0.5–7.7	[$mg \cdot L^{-1}$]	38	49	0.5–72
MBAS	[$g \cdot p^{-1} \cdot d^{-1}$]	5.6	4.4	2.5–8.7	[$mg \cdot L^{-1}$]	48	12	40–56
B	[$g \cdot p^{-1} \cdot d^{-1}$]	0.16	0.1	0–0.22	[$mg \cdot L^{-1}$]	0.6		
Na	[$g \cdot p^{-1} \cdot d^{-1}$]	12	14	2.0–32	[$mg \cdot L^{-1}$]	148	29	7.4–480

Taken from: Larsen, Udert and Lienert. Wastewater composition, Chapter 17. In: “Source separation and decentralisation for wastewater management”. IWA Publishing, London, 2013.

Some typical wastewater characteristics

Source	pH	TCOD mg/l	BOD_5 mg/l	TSS mg/l	FOG mg/l	TKN mg/l	TP mg/l	Other
Sewage	7	550	300	250	100	55	12	Pathogens
Petrochemical	7.8	650	250	100	20	13	3	
Textile Dyehouse	7.6	3000	1500	120	35	60	15	Colour
Monosodium Glutamate	3.2	50	35	10	-	8	300	SO_4^{2-} - 180 mg/l
Organic (benzene) chemicals	8.5	8000	3000	15	-	100	2	
Coke-ovens (steel manufacture)	9	2200	600	70	-	330	-	Phenols - 330 mg/l; SCN - 200 mg/l; CN - 100 mg/l
Oil refinery	6.5-9.5	1400	800	200	100	50	3	Phenol - 5 mg/l
Paper mill	7	10000	6000	150	-	30	-	
Yeast manufacturing	7	6500	2700	1100	-	260	-	TDS - 3000 mg/l Colour DON (220 mg/l)
Potato chip manufacture	7	15,000	9,000	7,000	100	180	25	Starchy wastewater
Potato frozen foods	6	10,000	5,000	4,500	200	250	110	Already undergone coarse screening and removal of excess fat
Fish processor	5.5	4,500	4,000	1,500	100	350	40	
Cheese	3.5	50,000	30,000	10,000	200	450	-	K^+ - 1300 mg/l; SO_4^{2-} - 300 mg/l
Meat processing	7	8000	5000	3500	2500	180	40	

Some typical wastewater characteristics

Source	pH	TCOD mg/l	BOD_5 mg/l	TSS mg/l	FOG mg/l	TKN mg/l	TP mg/l	Other
Sewage	7	550	300	250	100	55	12	Pathogens
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Organic (benzene) chemicals	8.5	8000	3000	15	-	100	2	
Coke-ovens (steel manufacture)	9	2200	600	70	-	330	-	Phenols - 330 mg/l; SCN - 200 mg/l; CN - 100 mg/l
Oil refinery	6.5-9.5	1400	800	200	100	50	3	Phenol - 5 mg/l
Paper mill	7	10000	6000	150	-	30	-	
Yeast manufacturing	7	6500	2700	1100	-	260	-	TDS - 3000 mg/l Colour DON (220 mg/l)
Potato chip	7	15,000	9,000	7,000	100	180	25	Starchy wastewater
							110	Already undergone coarse screening and removal of excess fat
							40	
							-	K^+ - 1300 mg/l; SO_4^{2-} - 300 mg/l
							40	

Manufacturing/chemical industry wastewater is typically:

- complex
- variable pH
- specific (refractory) compounds
- toxic/inhibitory compounds
- Low/BOD/COD ratio

Some typical wastewater characteristics

Source	pH	TCOD mg/l	BOD_5 mg/l	TSS mg/l	FOG mg/l	TKN mg/l	TP mg/l	Other
Sewage	7	550	300	250	100	55	12	Pathogens
Petrochemical	7.8	650	250	100	20	13	3	
Textile Dyehouse	7.6	3000	1500	120	35	60	15	Colour
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Paper mill	7	10000	6000	150	-	30	-	
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Potato chip manufacture	7	15,000	9,000	7,000	100	180	25	Starchy wastewater
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Some typical wastewater characteristics

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					100	50	3	Phenol - 5 mg/l
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Potato chip manufacture	7	15,000	9,000	7,000	100	180	25	Starchy wastewater
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Fish processor	5.5	4,500	4,000	1,500	100	350	40	
Cheese	3.5	50,000	30,000	10,000	200	450	-	K^+ - 1300 mg/l; SO_4^{2-} - 300 mg/l
Meat processing	7	8000	5000	3500	2500	180	40	

Food process wastewater is typically:

- High BOD
- Relatively high BOD/COD ratio
- high suspended solids and O&G
- relatively high in nutrients (N&P)

5.2 DO and BOD

- ❑ Oxygen solubility, Henry’s Law
- ❑ BOD5

Dissolved oxygen

Significance

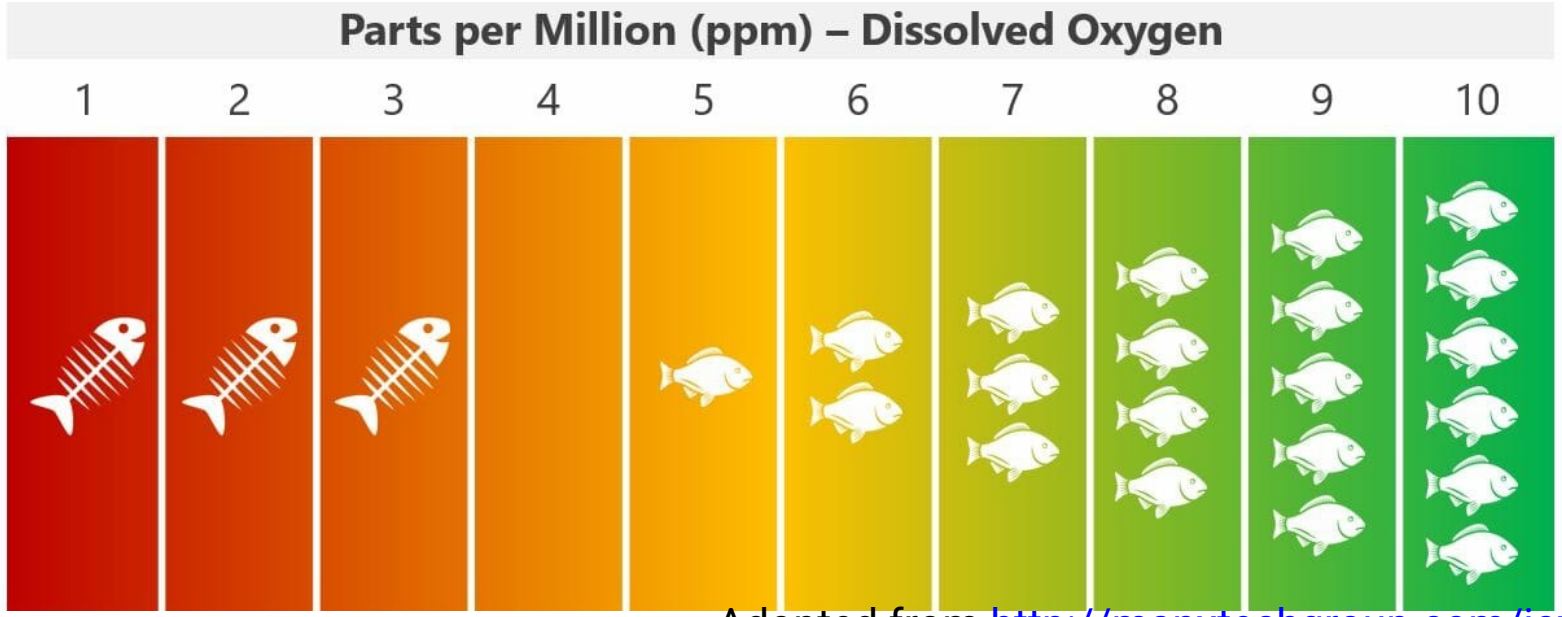
One of the most important water quality measure for rivers, streams, and lakes

Aquatic organisms need DO to survive

DO is required for aerobic biological treatment processes

Aerobic ⟸ O_2

Anaerobic ⟸ ✗ O_2



Adopted from http://manxtechgroup.com/iot

‘The smell is next level’: millions of dead fish spanning kilometres of Darling-Baaka river begin to rot

Releasing environmental water held downstream could help stem the spread of poor quality water, experts say

- **Follow our Australia news live blog for the latest updates**
- **Get our morning and afternoon news emails, free app or daily news podcast**



Drone footage shows millions of dead fish in river near Menindee - video

[The Guardian]

Environment

Low dissolved oxygen levels behind mass fish death in Australian river

Reuters

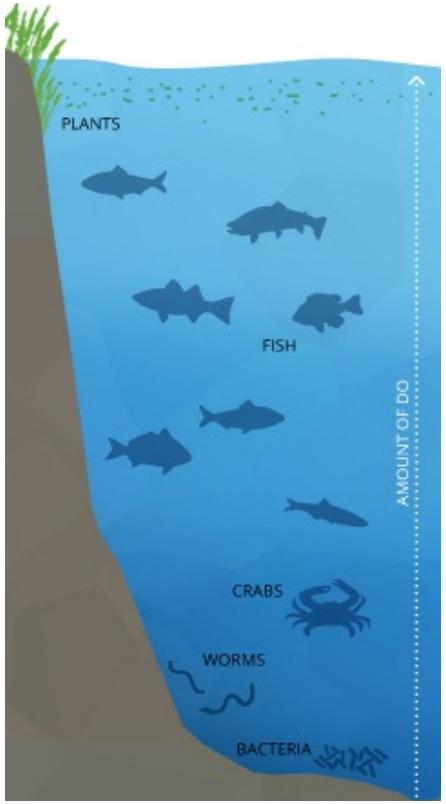
March 18, 2023 7:52 PM GMT+10 · Updated 2 months ago

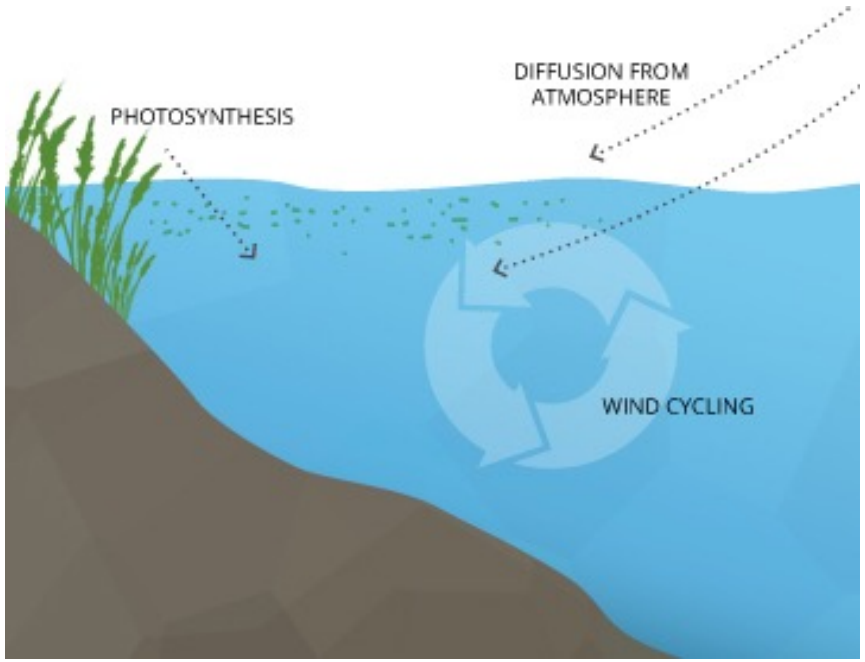


SYDNEY, March 18 (Reuters) - Low levels of oxygen in Australia's second longest river were to blame for a mass fish die-off recently in a remote part of New South Wales state, environmental authorities said.

[Reuters]

- Bottom feeders, crabs, oysters and worms need minimal amounts of oxygen (1-6 mg/L);
- Shallow water fish need higher levels (4-15 mg/L)



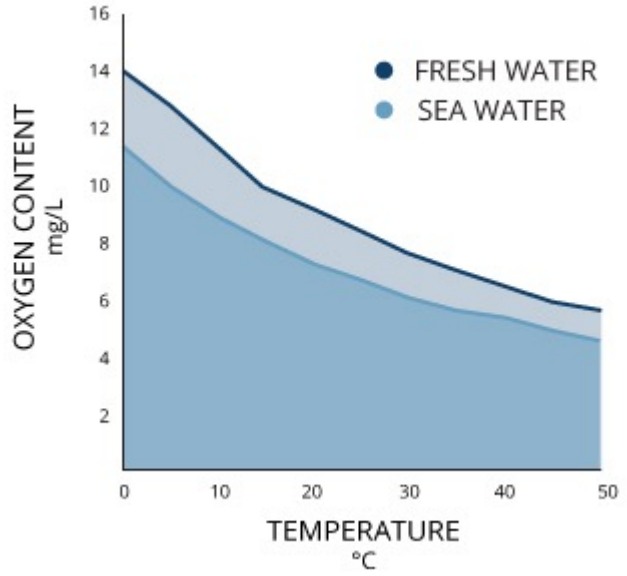


- From the **air**, oxygen can slowly diffuse across the water's surface from the surrounding atmosphere
- From the waste product of **photosynthesis** from phytoplankton, algae, seaweed and other aquatic plants

What Affects Oxygen Solubility?

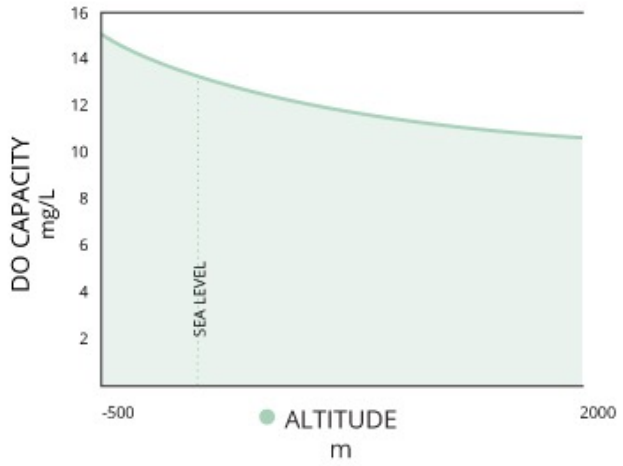
Two bodies of water that are both 100% air-saturated do not necessarily have the same concentration of dissolved oxygen.

The actual amount of dissolved oxygen (in mg/L) will vary depending on **temperature, pressure and salinity**.



- **Warmer** surface water requires less dissolved oxygen to reach 100% air saturation than does deeper, **cooler** water

- Dissolved oxygen decreases exponentially as **salt** levels increase



- Dissolved oxygen will increase as **pressure** increases

Dissolved oxygen

- Poorly soluble in water (~ 9 mg/L at 20 °C)
- Solubility depends on temperature, salinity and Pressure
  - Higher temperature → lower solubility
  - Higher salinity → lower solubility

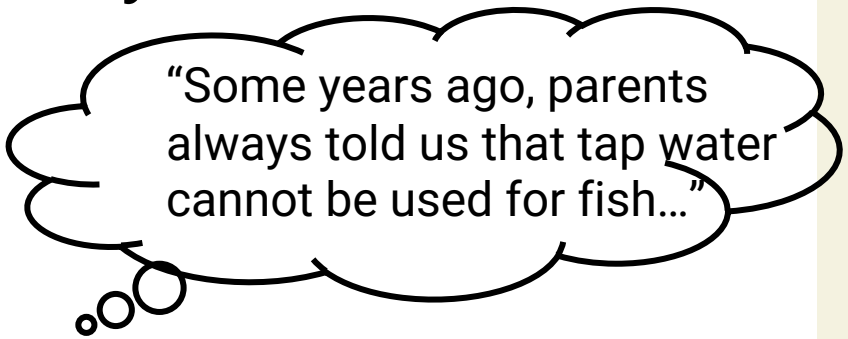


TABLE 5.10

Solubility of Oxygen in Water (mg/L) at 1 atm Pressure

Temperature (°C)	Chloride Concentration in Water (mg/L) 0	5,000	10,000	15,000
0	14.62	13.73	12.89	12.10
5	12.77	12.02	11.32	10.66
10	11.29	10.66	10.06	9.49
15	10.08	9.54	9.03	8.54
20	9.09	8.62	8.17	7.75
25	8.26	7.85	7.46	7.08
30	7.56	7.19	6.85	6.51

Source: Thomann and Mueller, 1987.

- $DO_s = K_H P_{O2}$

Textbook Table 5.10 (page 213)

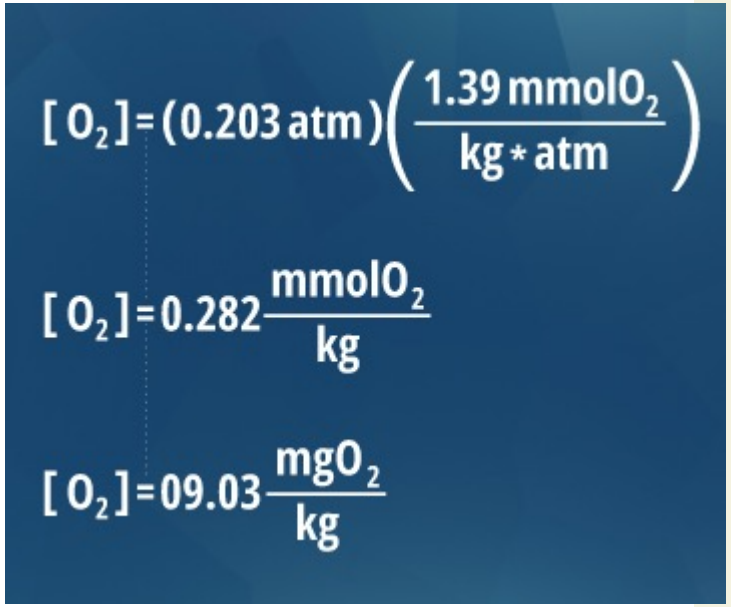
Henry's Law

$DO_s = K_H P_{O2}$

- K_H – Henry's Law Coefficient of O_2
- P_{O2} – The partial pressure of O_2

According to Henry's Law, the dissolved oxygen content of water is proportional to the percent of oxygen (partial pressure) in the air above it.

As oxygen in the atmosphere is about 20.3%, the partial pressure of oxygen at sea level (1 atm) is 0.203 atm. Thus the amount of dissolved oxygen at 100% saturation at sea level at 20 °C is 9.03 mg/L



Henry's law determining the dissolved oxygen concentration at 20 °C and 100% air saturation (1 kg water = 1 L water)

Temperature (°C)	Henry's Law Coefficient of O_2, K_H (mol/L·atm)
0	0.0021812
5	0.0019126
10	0.0016963
15	0.0015236
20	0.0013840
25	0.0012630

Example

By volume, the concentration of oxygen in air is about 21%. Find the saturated DO value (i.e., DO_s) in water (in mol/L and mg/L) at 25 °C and 1 atmosphere of pressure.

Solution:

Air is 21% O_2, so its partial pressure (i.e, P_{O2}) at 1 atm is: $P_{O2} = 0.21 \times 1$ atm = 0.21 atm.

From above table, at 25 °C, K_H = 0.0012630 mol/L·atm; so, according to the Henry's Law:

$DO_s = K_H P_{O2}$ = 0.0012630 mol/L·atm × 0.21 atm = **2.65×10^{-4} mol/L**

The molecular weight of O_2 is 32 g/mol, so the DO_s = 2.65×10^{-4} mol/L × 32×10^3 mg/mol = **8.5 mg/L**

Or 8.5 ppm

Dissolved oxygen

- Oxygen demanding **(OD)** waste
  - Organic matter such as proteins, carbohydrates, fats, etc.
  - Some inorganic matter
  - reduced DO

Typical Domestic Wastewater Organics

		% biodegradable
ORGANICS -	Proteins (40-60%)	40-50%
	Carbohydrates (30-50%)	75-80%
	Fats (10%)	80-85%

Other organics:
- Synthetic detergents
- Organic solvents
- Petroleum products
- Biological polymers
- Phenols
- Pesticides

From Stanford CEE271B Course Notes, C. Criddle

Biochemical oxygen demand

The quantity of oxygen required by microorganisms to oxidise organic wastes aerobically into stable products is called the **biochemical oxygen demand (BOD, or biological oxygen demand).**

$$\text{Organic matter} + O_2 \xrightarrow{\textit{microorganisms}} CO_2 + H_2O + \text{new cells} + \text{stable products } (SO_4^{2-}, PO_4^{3-}, NO_3^{-})$$

BOD test is used to determine the strength of oxygen demanding waste

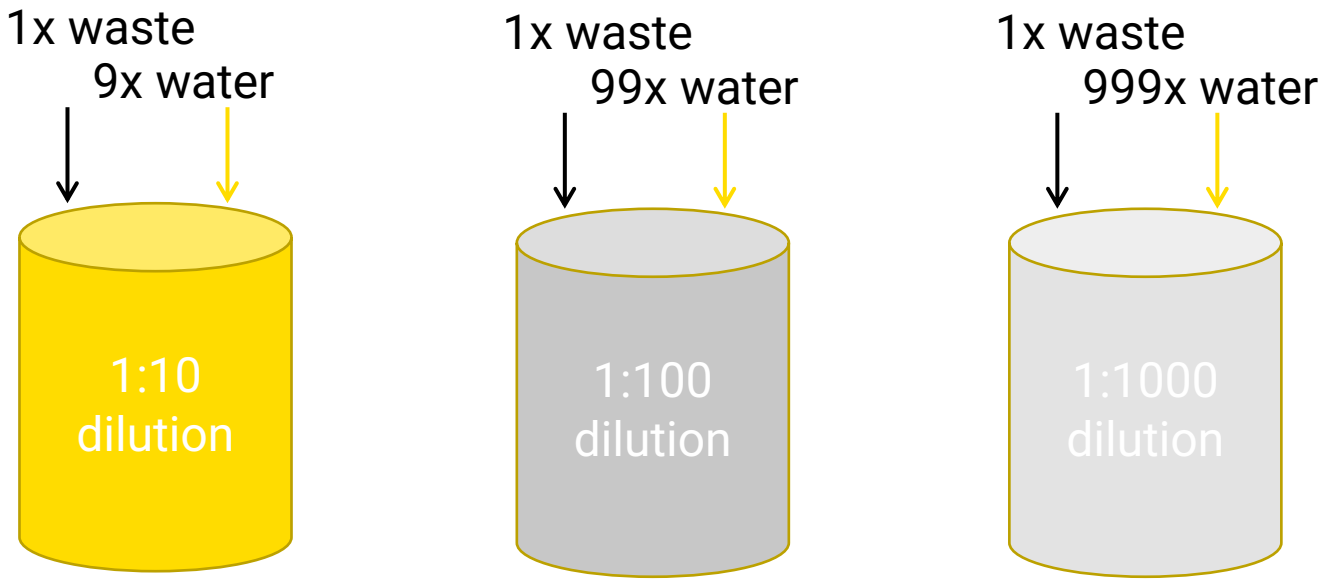
An important measure of the impact of the waste on receiving water (e.g., discharge of wastewater into a river).

The BOD value is most **commonly expressed in milligrams of oxygen consumed per liter of sample during 5 days of incubation at 20 °C** and is often used as a surrogate of the degree of organic water pollution.



5-day BOD test



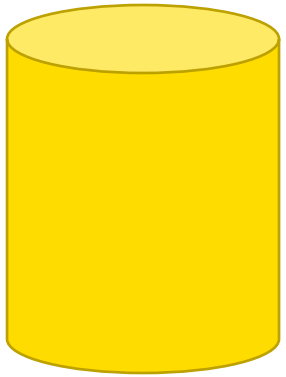


Standard 300 mL BOD bottle

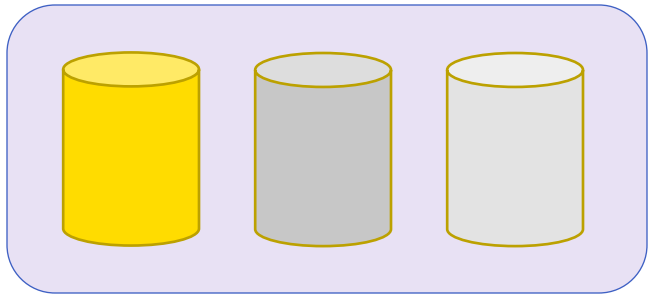
Series of dilution

Temperature: 20°C ± 1°C in an air incubator or water bath

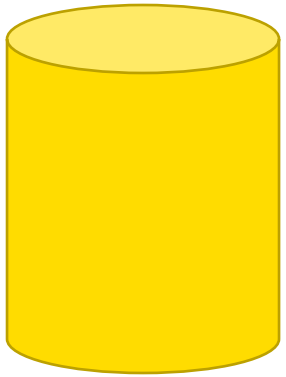
5-day BOD test



Initial DO measurement (DO_i)



Incubation at 20°C ± 1°C for **5** days



Final DO measurement (DO_f)

$$BOD_5 = \frac{DO_i - DO_f}{P}$$

P -- Decimal volumetric fraction of sample used (Pollutants diluted, the fraction of the mixture) = V_w/V_m

Example 5.1 Unseeded Five-Day BOD test

A 10 mL sample of sewage mixed with enough pure water to fill a standard 300 mL BOD bottle has an initial DO of 9.0 mg/L. To help assure an accurate test, it is desirable to have at least a **2.0 mg/L drop** in DO during the five-day test, and **the final DO should be at least 2.0 mg/L**. For what range of BOD_5 would this dilution produce the desired results?

Solution:

The dilution fraction is P = 10/300. To get at least a 2.0 mg/L drop in DO, the minimum BOD needs to be:

$BOD_5 \geq (DO_i - DO_f)/P$ = 2.0 mg/L/(10/300) = **60 mg/L**

To assure at least 2.0 mg/L of DO remains after five-day requires that:

$BOD_5 \leq (DO_i - DO_f)/P$ = (9.0-2.0) mg/L/(10/300) = **210 mg/L**

So the dilution will be satisfactory for BOD_5 values **between 60 and 210 mg/L**

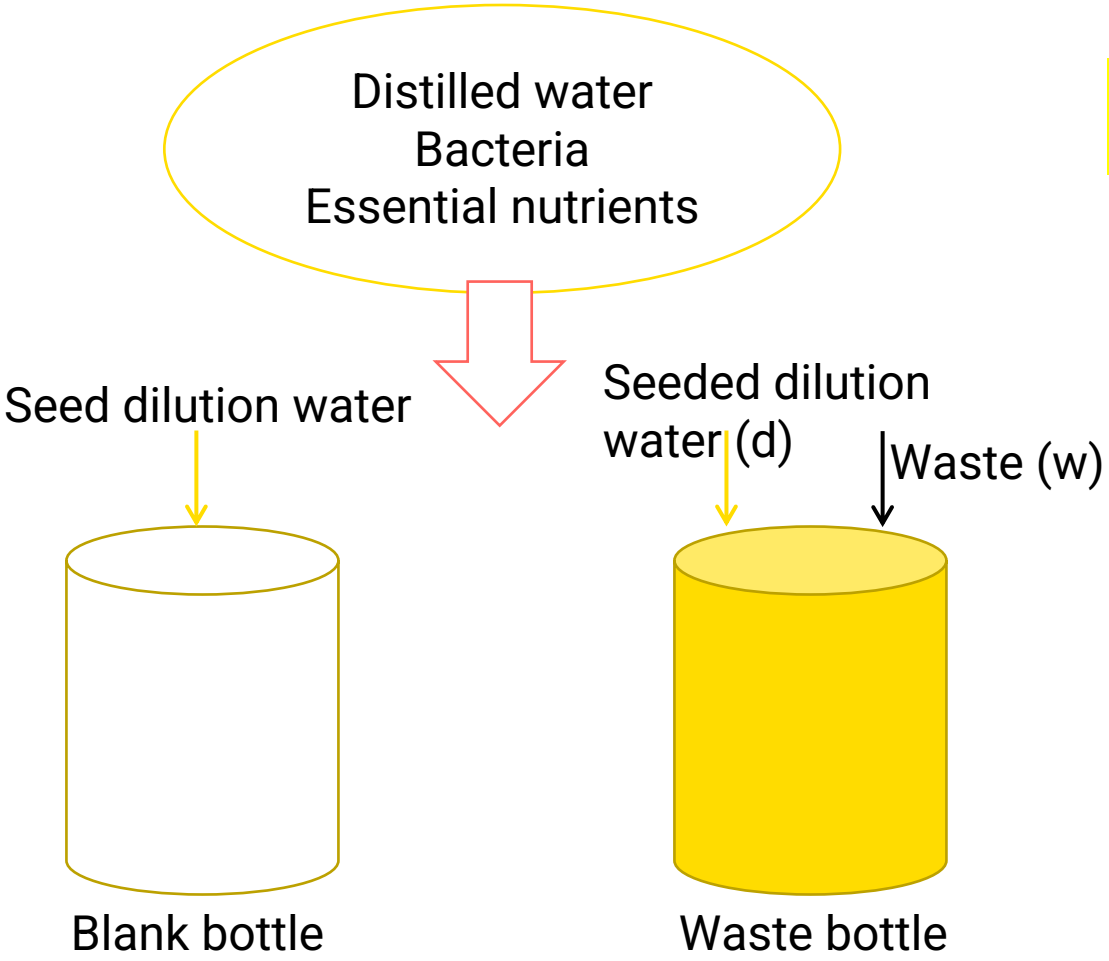
Notes

So far, we have assumed that the dilution water added to the wastewater sample ***has no BOD of its own***, which would be the case if pure water were added. In some cases it is necessary to seed the dilution water with microorganisms ***to assure that there is an adequate bacterial population to carry out the biodegradation***. In such cases, to find the BOD of the wastewater itself, it is necessary to subtract the oxygen demand caused by the seed from the demand in the mixed sample of wastewater and dilution water.

5.3 BOD

- ❑ **Ultimate BOD, BOD modeling**, and temperature correction
- ❑ CBOD vs. NBOD
- ❑ BOD vs. other parameters for indicating organic waste strength

Seeding and blank test



$$M_m = M_w + M_d$$

$$BOD_m V_m = BOD_w V_w + BOD_d V_d$$

$$BOD_5 \text{ as } mgO_2 / L = \frac{(DO_i - DO_f) - (B_i - B_f)(1 - P)}{P}$$

where:

DO_i = initial DO of sample, mg/L

DO_f = Final DO of incubated sample after 5 days, mg/L

B_i = DO of seed control before incubation, mg/L

B_f = DO of seed control after incubation, mg/L

P = Decimal volumetric fraction of sample used

Calculation of seeded BOD

$$BOD_m V_m = BOD_w V_w + BOD_d V_d$$

BOD_m = BOD of the mixture of wastewater and seeded dilution water
BOD_w = BOD of the wastewater alone
BOD_d = BOD of the seeded dilution water alone (i.e., the blank)
V_w = the volume of wastewater in the mixture
V_d = the volume of seeded dilution water in the mixture
V_m = the volume of the mixture = V_w + V_d

Calculation of seeded BOD

Let *P* = the fraction of the mixture that is wastewater = V_w/V_m so that (1-*P*) = the fraction of the mixture that is seeded dilution water = V_d/V_m. Rearranging the equation as:

$$BOD_w = BOD_m(V_m/V_w) - BOD_d(V_d/V_w \times V_m/V_m)$$

where the last term has been multiplied by unity (V_m/V_m). A slight rearrangement of the equation as:

$$BOD_w = BOD_m(V_m/V_w) - BOD_d(V_d/V_m)/(V_w/V_m)$$

Substituting the definition of *P* and (1-*P*) into the above equation:

$$BOD_w = [BOD_m - BOD_d(1-P)]/P$$

Because $\mathbf{BOD_m = DO_i - DO_f}$ and $\mathbf{BOD_d = B_i - B_f}$; $\mathbf{B_i}$ **= initial DO in the seeded dilution water**, and $\mathbf{B_f}$ **= final DO in the seeded dilution water**. So,

$$\mathbf{BOD_w} = \frac{(\mathbf{DO_i - DO_f}) - (\mathbf{B_i - B_f})(\mathbf{1 - P})}{\boldsymbol{P}}$$

Calculation of seeded BOD

Example 5.2 Seeded BOD

A test bottle containing just seeded dilution water has its DO level drop by 1.0 mg/L in a five-day test. A 300 mL BOD bottle filled with 15 mL of wastewater and the rest seeded dilution water (sometimes expressed as a dilution of 1:20) experiences a **drop of 7.2 mg/L** in the same period. What would be the BOD_5 of the wastewater?

Solution:

The dilution fraction is P = 15/300 = 0.05

$BOD_m = DO_i - DO_f$ = 7.2 mg/L, $BOD_d = B_i - B_f$ = 1.0 mg/L

So:

$$\mathbf{BOD_w} = \frac{(\mathbf{DO_i - DO_f}) - (\mathbf{B_i - B_f})(\mathbf{1 - P})}{\boldsymbol{P}} = \frac{\mathbf{7.2 - 1.0 \times (1 - 0.05)}}{\mathbf{0.05}} = \mathbf{125}\ \boldsymbol{mg/L}$$

Modeling BOD

Imagine a flask with some biodegradable organic waste in it.

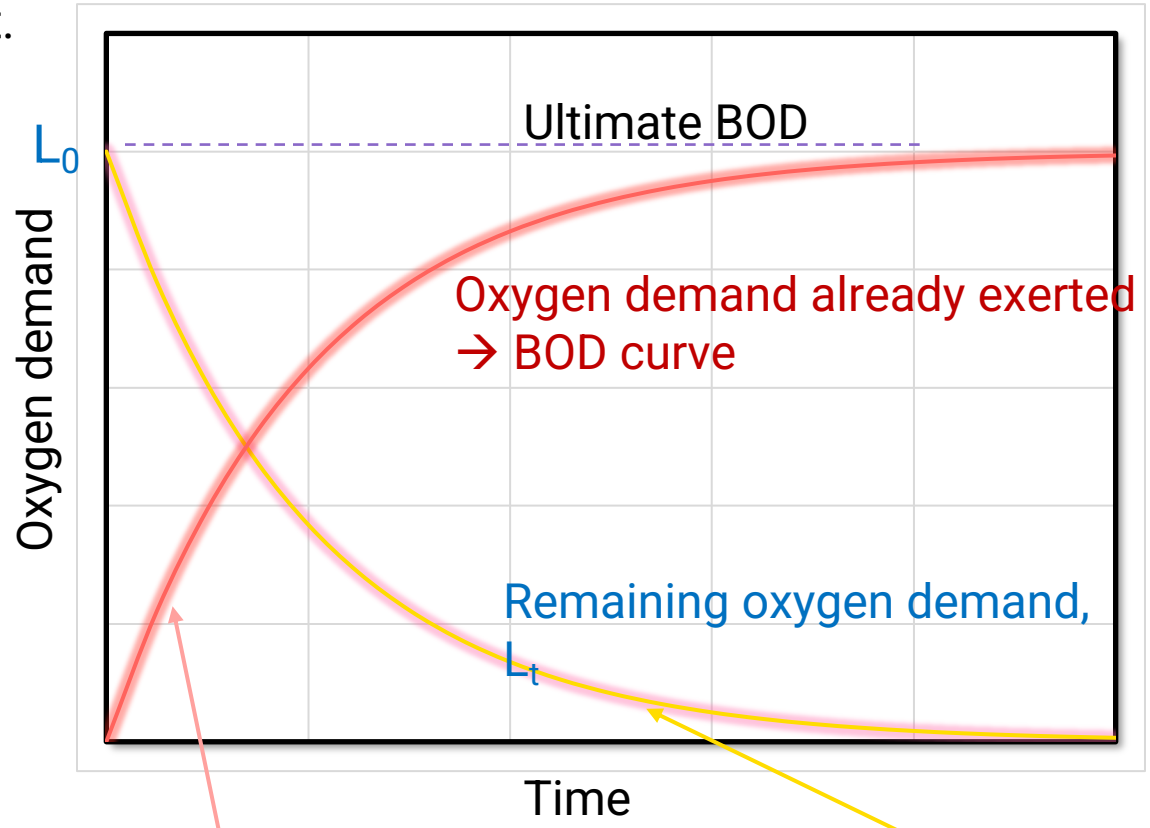


Figure. Two equivalent ways to describe the time dependence of organic matter in a flask.

First order reaction:

$$\frac{dL_t}{dt} = -kL_t$$

Calculus

$$L_t = L_0 e^{-kt}$$

$$BOD_t = L_0\,(1 - e^{-kt})$$

$$BOD_t = L_0\,(1 - 10^{-Kt})$$

$$k = 2.3K$$

The amount of organic matter remaining in the flask will decrease with time until it all disappears.

The amount of organic matter already oxidized goes up until finally all of the original organic matter has been oxidized as time goes on.

First order reaction:

$$\frac{dL_t}{dt} = -kL_t$$

The rate of decomposition of organic wastes

Calculus

k – the BOD reaction constant, proportional value; L_t – the amount of waste that is left in the flask.

$$L_t = L_0 e^{-kt}$$

L_0 – the ultimate carbonaceous oxygen demand, total amount of oxygen required by microorganisms.

$L_0 = BOD_t + L_t$

$$BOD_t = L_0 (1 - e^{-kt})$$

$$BOD_t = L_0 (1 - 10^{-Kt})$$

$$k = 2.3K$$

Modeling BOD Temperature correction

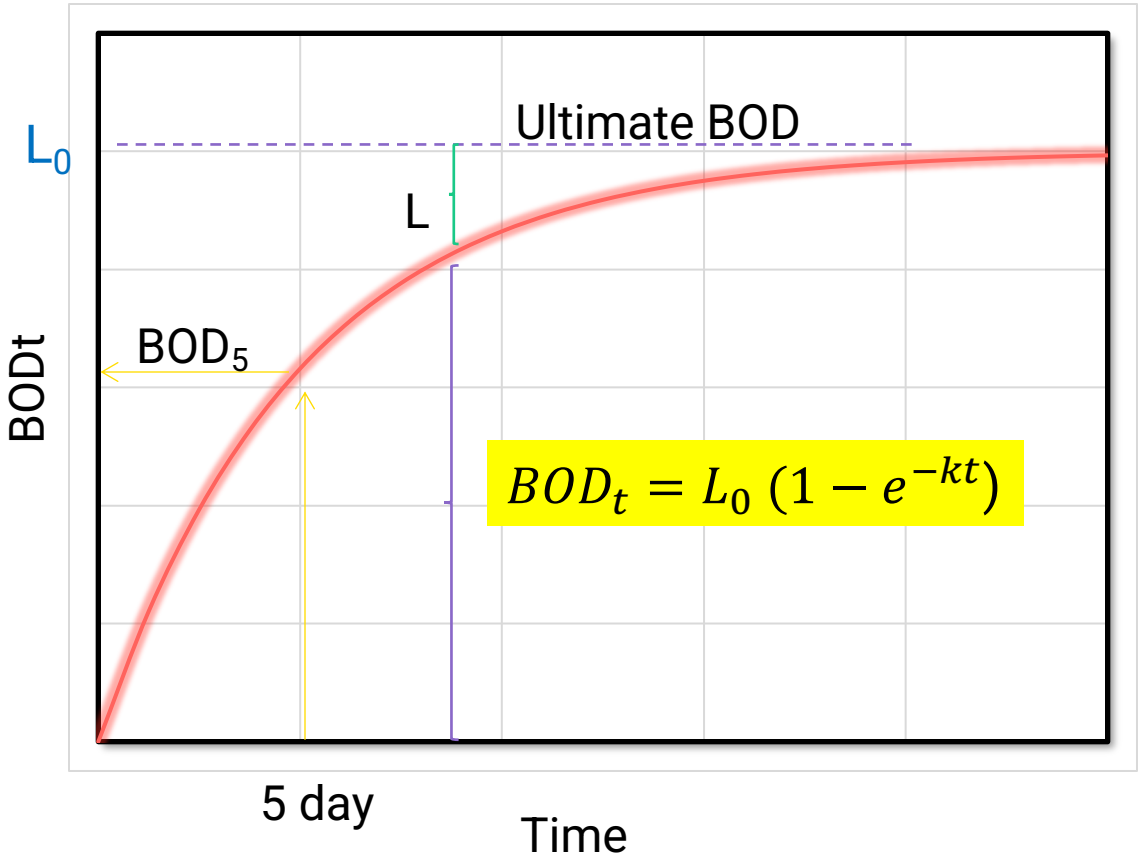


Figure. Idealized carbonaceous oxygen demand, the oxygen already consumed as a function of time.

k – the BOD reaction constant, a constant value. It is factor that indicates the rate of biodegradation of wastes.

The reaction rate will depend on a number of factors, including the nature of waste itself.

Temperature correction:

$$k_T = k_{20}\theta^{T-20} \text{ where } \theta = 1.047$$

k_{20} -- the reaction rate constant at the standard 20 °C laboratory reference temperature.

Example 5.3 Estimating L_0 from BOD_5

The dilution factor *P* for an unseeded mixture of wastewater and pure water is 0.030. The DO of the mixture is initially 9.0 mg/L, and after five days, it has dropped to 3.0 mg/L. The reaction rate constant *k* has been found to be 0.22 day^{-1}. Questions:

a. What is the five-day BOD of the waste?
b. What would be the ultimate BOD?
c. What would be the remaining oxygen demand after five days?

Solutions:

a. $$\mathrm{BOD}_5 = \frac{\mathrm{DO_i} - \mathrm{DO_f}}{P} = \frac{9.0 - 3.0}{0.030} = \mathbf{200\ mg/L}$$

b. According to the equation, the ultimate carbonaceous BOD (i.e., L_0) is:

$$L_0 = \frac{\mathrm{BOD}_5}{1 - e^{-kt}} = \frac{200}{1 - e^{-0.22\times5}} = \mathbf{300\ mg/L}$$

c. After five days, 200 mg/L of oxygen demand out of the total 300 mg/L would have already been used. The remaining oxygen demand would therefore be:

$$L_t = L_0 - \mathrm{BOD}_5 = 300 - 200 = \mathbf{100\ mg/L}$$

Example 5.4 Temperature dependence of BOD_5

In the previous example, the wastewater had an L_0 of 300 mg/L. At 20 °C, the BOD_5 was 200 mg/L, and the reaction rate constant (i.e., k) was 0.22 day^{-1}. What would the of this wastewater be at 25 °C.

Solution:

Adjusting the k value using temperature correction:

$$k_{25} = k_{20} 1.047^{T-20} = 0.22 \times 1.047^{25-20} = 0.277 \text{ day}^{-1}$$

So, the BOD_5 at 25 °C should be:

$$\text{BOD}_5 = L_0\left(1 - e^{-kt}\right) = 300 \times (1 - e^{-0.277 \times 5}) = \mathbf{225\ mg/L}$$

Notice that BOD_5 at 25 °C is somewhat higher than the value at 20 °C. The same amount of oxygen is required at either temperature, but as temperature increases, it gets used sooner.

Typical Values for the BOD Rate Constant at 20 °C

Sample	k (day^{-1})	K (day^{-1})
Raw sewage	0.35-0.70	0.15-0.30
Well-treated sewage	0.12-0.23	0.05-0.10
Polluted river water	0.12-0.23	0.05-0.10

Source: Davis and Cornwell, 1991.

Measures of oxygen demand

Carbonaceous BOD (CBOD) ⇨ BOD_5, L_0 ← OD -- Only associated with the biodegradation of the carbonaceous portion of the wastes!

- Oxygen demand for oxidation of organic carbon

$$\text{Organic matter} + O_2 \rightarrow CO_2 + H_2O + NH_3$$

Nitrogenous BOD (NBOD) ← OD -- Associated with the biodegradation of the nitrogen compounds.

- Oxygen demand for oxidation of ammonia to nitrate

$$NH_3 + 2O_2 \rightarrow H^+ + NO_3^- + H_2O$$

$$2NH_3 + 3O_2 \xrightarrow{\text{Nitrosomonas}} 2H^+ + 2NO_2^- + H_2O$$

$$2NO_2^- + O_2 \xrightarrow{\text{Nitrobacter}} 3NO_2^-$$

Nitrification

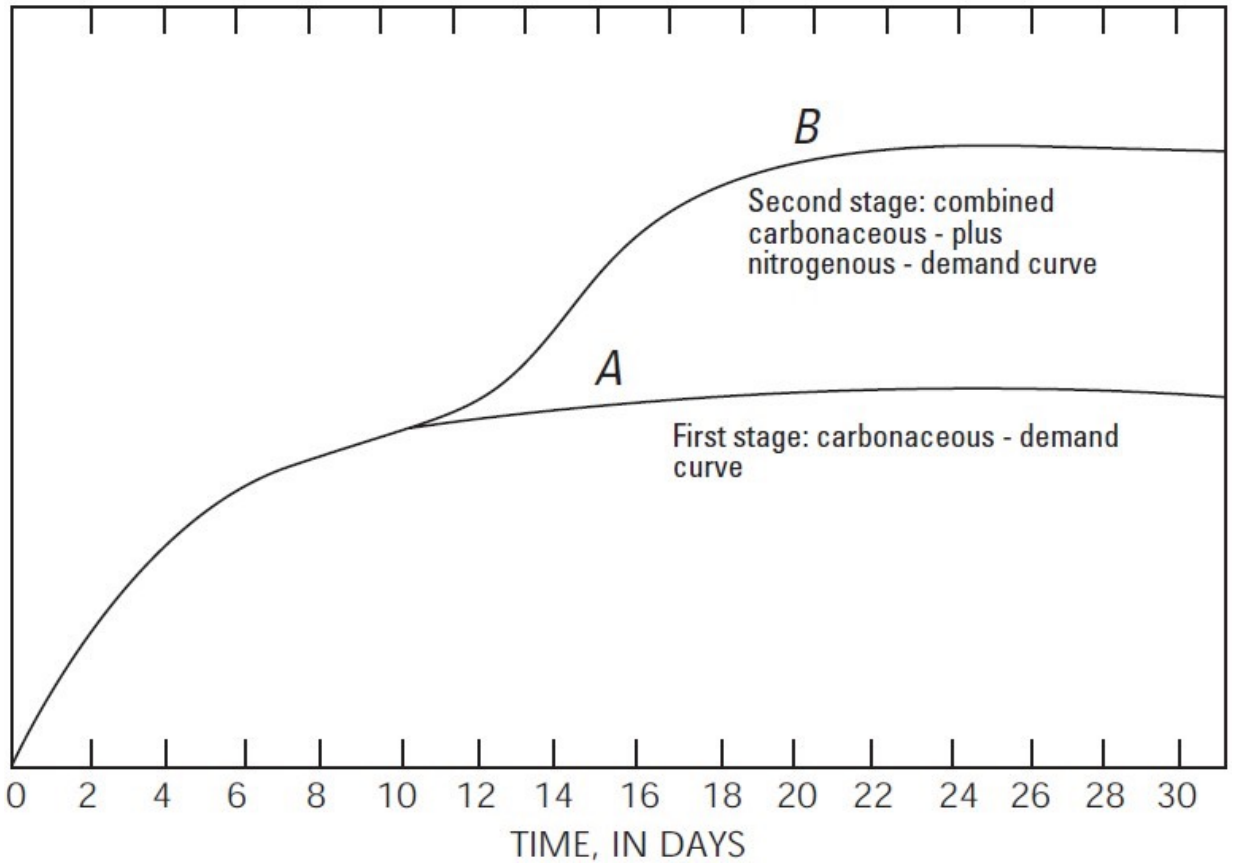
Total BOD

- Oxygen demand for oxidation of organic carbon

$$\text{Organic matter} + O_2 \rightarrow CO_2 + H_2O + H^+ + NO_3^-$$

Total BOD = CBOD + NBOD

CBOD and NBOD



Illustrating the carbonaceous and nitrogenous biochemical oxygen demand. The BOD is the sum of the two.

CBOD and NBOD

L_0 – the ultimate carbonaceous oxygen demand, total amount of oxygen required by microorganisms for carbonaceous.

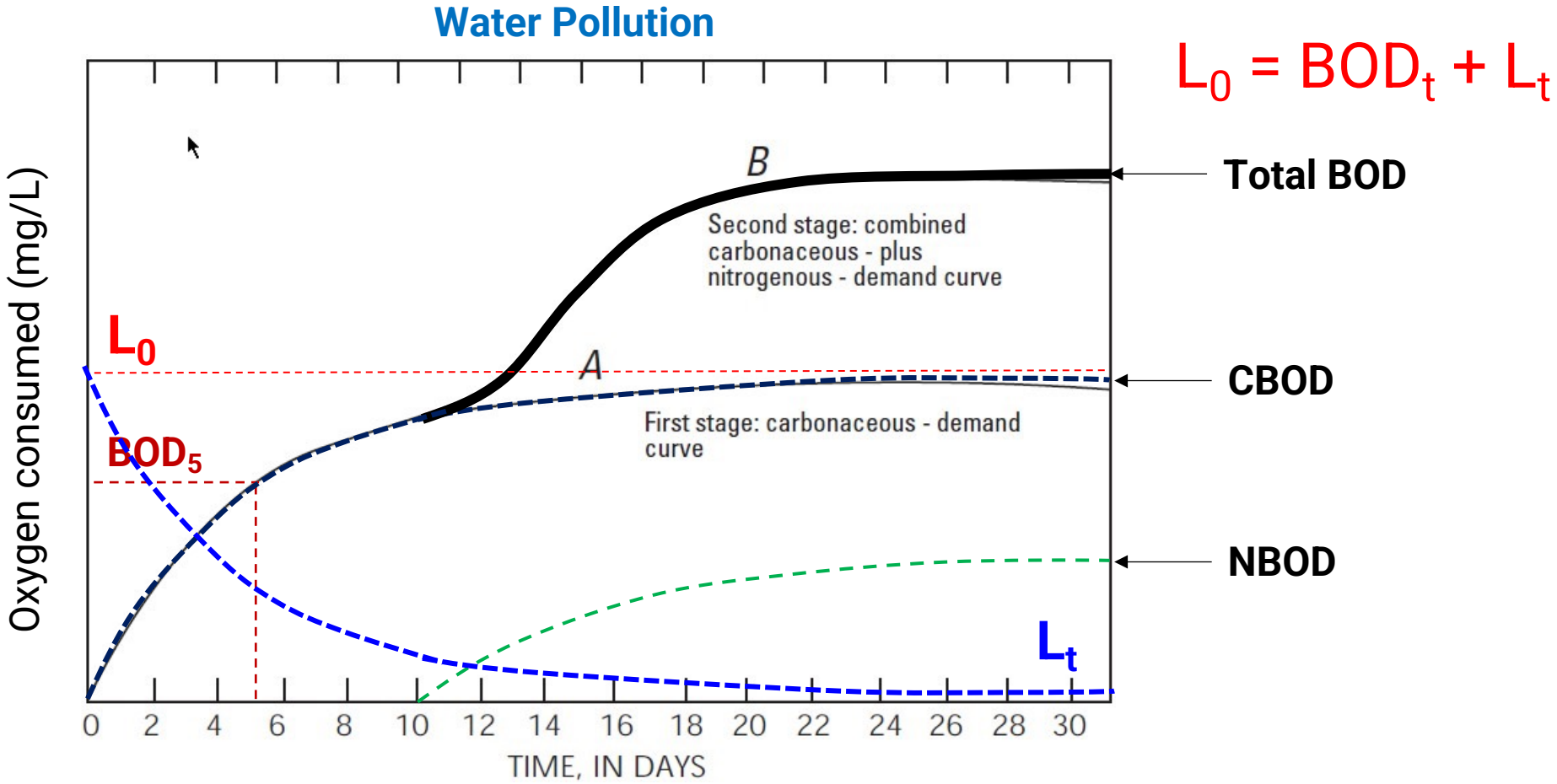


Fig. Illustrating the carbonaceous and nitrogenous biochemical oxygen demand.

Example 5.5 Nitrogen oxygen demand

Some domestic wastewater has 30 mg/L of nitrogen in the form of either organic nitrogen or ammonia. Assuming that very few cells of bacteria are formed during the nitrification of the wastewater (i.e., the oxygen demand can be found from a simple stoichiometric analysis of the nitrification reactions). Questions:

a. The ultimate nitrogenous oxygen demand.
b. The ratio of the ultimate NBOD to the concentration of nitrogen in the wastewater.

Solutions:

a. $NH_3 + 2O_2 \rightarrow NO_3^- + H^+ + H_2O$ (or $NH_3 + 2O_2 = NO_3^- + H^+ + H_2O$)

1 mol NH_3 consume 2 mol O_2 in the reaction, 30 mg/L N in the wastewater (i.e., 30/14 = 30/14 = 2.14 mmol), so the O_2 demand (i.e., NBOD) should be 2.14 × 2 × 32 = ~ **137 mg/L**.

b. The oxygen demand due to nitrification divided by the concentration of nitrogen in the wastewater is:

$$\frac{137 \text{ mg } O_2/\text{L}}{30 \text{ mg N/L}} = \mathbf{4.57 \text{ mg } O_2/\text{mgN}}$$

Aggregate measures of organics in wastewater

Major parameters used to indicate organic waste strength include:

- Biochemical Oxygen Demand **BOD** ➡ **Biological**
- Chemical Oxygen Demand **COD** ➡ **Chemically**
- Total Organic Carbon **TOC**
- Theoretical Oxygen Demand **ThOD**

Chemical oxygen demand

Amount of oxygen required to **chemically oxidize** the waste. The chemical used is potassium dichromate ($K_2Cr_2O_7$, a strong oxidizing agent under acidic conditions).

$$C_nH_aO_bN_c + dCr_2O_7^{2-} + (8d + c)H^+ \longrightarrow nCO_2 + \frac{a + 8d - 3c}{2}H_2O + cNH_4^+ + 2dCr^{3+}$$

Does not depend on the ability of microbes to degrade the waste

- Quick measurement (a few hours): Easily quantifying the amount of organics in water.
- Biodegradable organics + "inert" organics: Quantifying the amount of oxidizable pollutants found in surface water or wastewater.
- No information on the rate of biodegradation
- The ratio of BOD/COD → treatability of a waste by biological treatment

Approximation: COD = CBOD, Assuming: No oxidizable inorganic substances ; Organic constituents are not toxic in BOD measurements. In reality, COD > CBOD

Total organic carbon (TOC)

- The **total amount of carbon from organic material** present in a sample;

- The advantages of TOC analysis are the fast analysis time of a few minutes, the exact and matrix-independent quantification and the very low consumption of chemicals.

- Total carbon (TC) = total organic carbon (TOC) + total inorganic carbon (TIC)

  TIC representing the amount of non-organic carbon, like carbon in carbonate minerals.

Theoretical oxygen demand (ThOD)

- The calculated amount of oxygen required to oxidize a compound to its final oxidation products.

Determining the Theoretical Oxygen Demand (ThOD)

- For a known contaminant, the ThOD can be calculated simply from the stoichiometric of the oxidation reaction involved.
- For example, what is the ThOD of a solution containing 200 mg/L of glucose ($C_6H_{12}O_6$)?
- Glucose oxidation can be expressed with the following chemical reaction

$$C_6H_{12}O_6 + O_2 \rightarrow CO_2 + H_2O$$

- Then the reaction needs to be balanced

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$$

- Considering the molecular weights for glucose and O_2 are 180 g/mol and 32 g/mol respectively, we can now determine the ThOD as:

$$\frac{200\ mg\ gluc}{L} \times \frac{mol\ gluc}{180\ g\ gluc} \times \frac{6\ mol\ O2}{mol\ gluc} \times \frac{32\ g\ O2}{mol\ O2} = 214 \frac{mg\ O2}{L} = \mathbf{214} \frac{\boldsymbol{mg\ COD}}{\boldsymbol{L}}$$

What do you think the BOD would be?

Examples in past examination

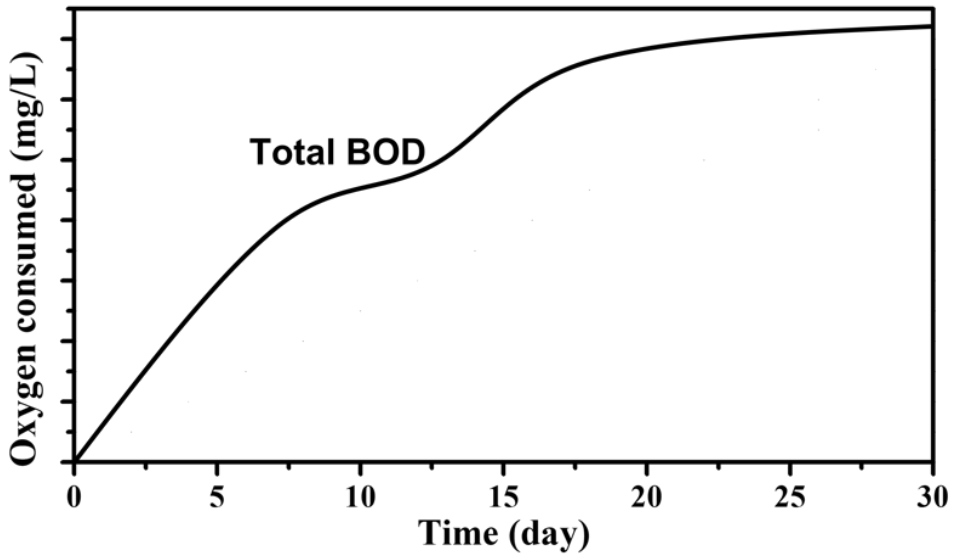
1. Which of the following pollutants is NOT oxygen demanding organic waste?
(a) phenol, (b) protein, (c) nitrate, or (d) pesticide.

2. Which of the following does NOT describe COD test?
(a) it is not suitable for samples containing toxic substances,
(b) it can be completed within 3 hours,
(c) it can not provide information of biodegradation rate, or
(d) it uses strong oxidants.

Exam 12-2023

The following picture illustrates the carbonaceous and nitrogenous biochemical oxygen demand. Please complete the following picture:
Please draw out the CBOD, NBOD, and the remaining OD curves in the following picture;
Please point out the BOD_5 in this picture;
Please explain CBOD, NBOD, total BOD, and the ultimate carbonaceous oxygen demand L_0.

(8 marks)



(b) The dilution factor for an unseeded mixture of wastewater and pure water is 0.08. The DO of the mixture was initially 10.0 mg/L, and 5 days later, it dropped 8.0 mg/L. The reaction rate constant k has been found to be 0.22 day^{-1}. What would be the ultimate CBOD?

(8 marks)

5.4 Water quality evaluation

❑ DO in rivers

----Oxygen sag curve

The **oxygen sag curve** describes how **DO** in a river changes downstream after wastewater (or any oxygen-demanding pollution) is discharged.

Oxygen-Demanding

I want to tell you:



1. **Organisms demanding**
2. **Oxygen Replenishment**



Effect of wastes on DO in rivers

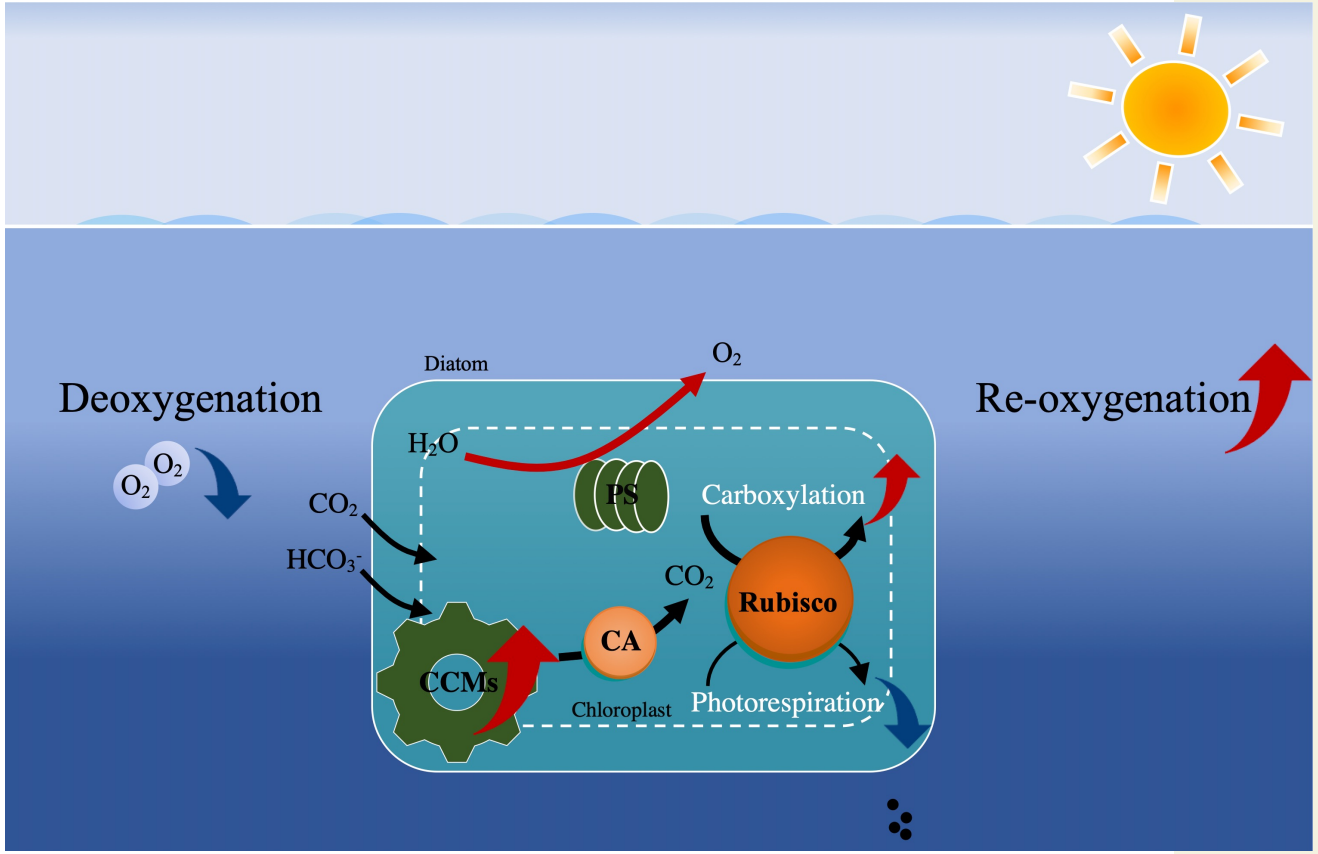
Two key processes of the simplest model of oxygen resources in a river:

1) The removal of oxygen by microorganisms during biodegradation;

Deoxygenation

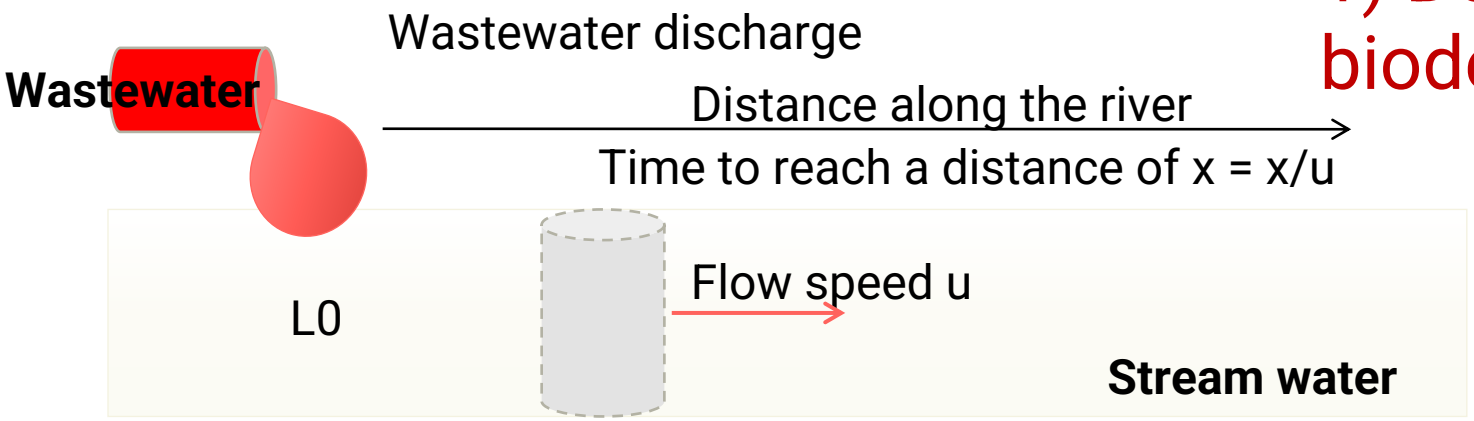
2) The replenishment of oxygen through reaeration at the interface river and the atmosphere.

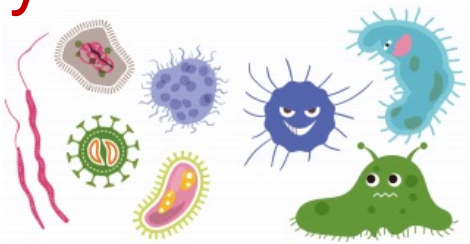
Reoxygenation (Reaeration)



Effect of wastes on DO in rivers

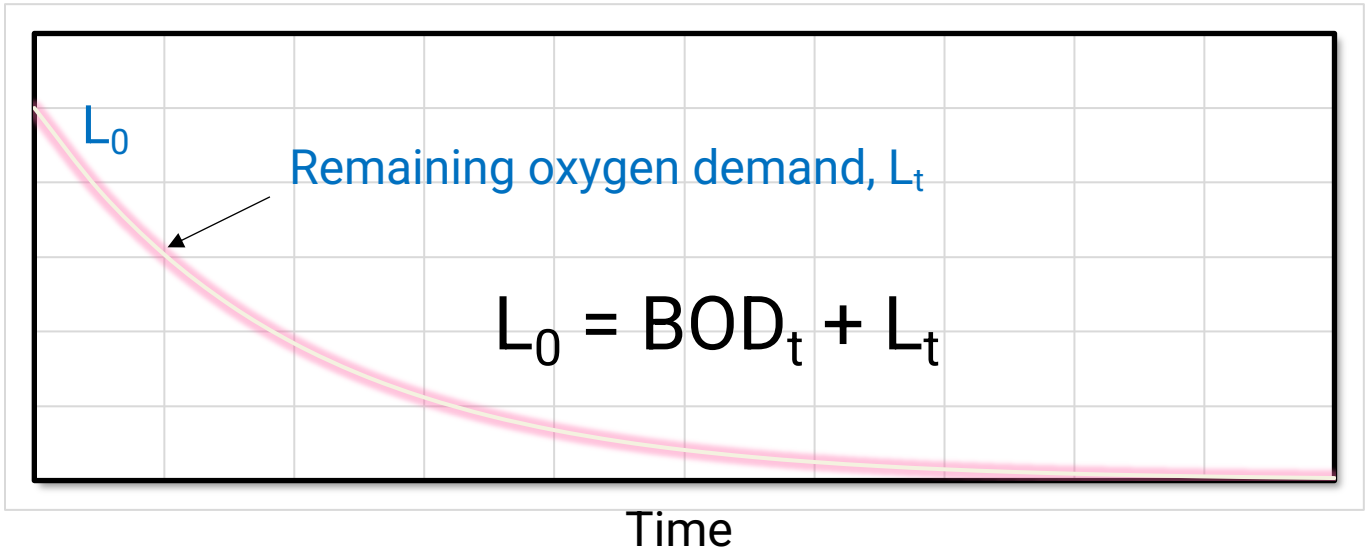
1) **Deoxygenation** by biodegradation





De-oxygenation (oxygen removal due to biodegradation):

Rate of deoxygenation = $k_d L_t$



K_d – the deoxygenation rate constant (day^{-1}), also, the BOD reaction constant (in a stand lab BOD test);

L_t – the BOD remaining t (days) after the wastes enter the river, (mg/L).

$$L_t = L_0 e^{-kt}$$

Rate of deoxygenation = $k_d\, L_0 e^{-kt}$

In a standard laboratory BOD test,
$k = k_d$

Rate of deoxygenation = $k_d\, L_0 e^{-kdt}$

In river, L_0 is the BOD of the mixture of stream water and wastewater (mg/L)

$\mathrm{BOD_m V_m = BOD_w V_w + BOD_d V_d}$ mass balance

$$L_0\big(V(waste) + V(river)\big) = BOD(waste)V(waste) + BOD(river)V(river)$$

V = Q t V– volume; t – time; Q – volumetric flow rate mg/L

$$L_0(Q_\mathrm{w} + Q_r) = L_\mathrm{w} Q_\mathrm{w} + L_\mathrm{r} Q_r$$

Example 5.6 Downstream BOD

A wastewater treatment plant serving a city discharges 1.10 m^3/s of treated effluent having an ultimate BOD of 50.0 mg/L into a stream that has a flow of 8.70 m^3/s and a BOD of its own equal to 6.0 mg/L. The deoxygenation constant k_d is 0.20/day.

a. Assuming complete and instantaneous mixing, estimate the ultimate BOD of the river just downstream from the outfall.
b. If the stream has a constant cross section, so that it flows at a fixed speed equal to 0.30 m/s, estimate the BOD remaining in the stream at a distance 30,000 m downstream.

Solutions:

a. The BOD of the mixture of effluent and streamwater can be found as:

$$L_0 = \frac{Q_w L_w + Q_r L_r}{Q_w + Q_r} = \frac{1.10 \times 50.0 + 8.70 \times 6.0}{1.10 + 8.70} = \mathbf{10.9\ mg/L}$$

b. At a speed of 0.30 m/s, the time required for the waste to reach a distance 30,000 m downstream would be:

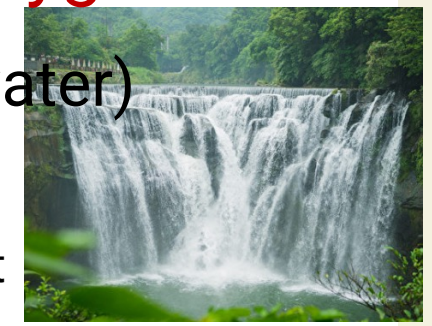
$$t = \frac{x}{u} = \frac{30{,}000}{0.30} = 100{,}000\ \text{s} = 1.16\ \text{days}$$

So the BOD remaining at that point, 30 km downstream, would be:

$$L_t = L_0 e^{-k_d t} = 10.9 \times e^{-0.20 \times 1.16} = \mathbf{8.7\ mg/L}$$

Effect of wastes on DO in rivers

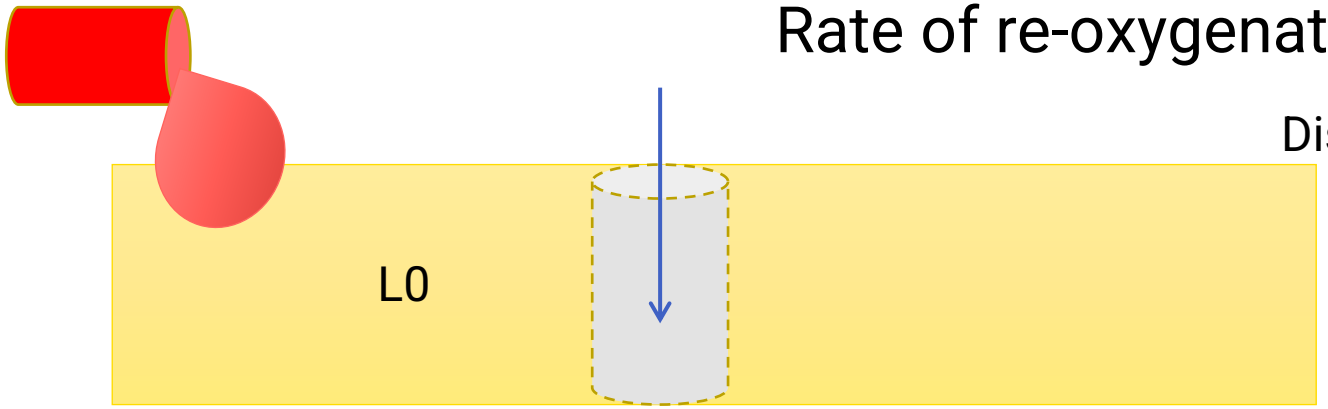
2) **Reaeration** for Oxygenation



Re-aeration (transfer of oxygen from air to water)

Rate of re-oxygenation = $k_r(DO_s - DO)$

Dissolved Oxygen gradient



k_r – reaeration constant (time^{-1})
DO_s – Saturated value of dissolved oxygen
DO – actual dissolved oxygen at a given location downstream

DO_s :

- saturated value of DO
- Determined by the Henry's law

k_r

- enhanced mass transfer for faster moving waters
- Empirical formula $k_r = \frac{3.9u^{1/2}}{H^{3/2}}$

u – average stream velocity (m/s);
H – average stream depth (m).

In river, DO_0 is the initial **oxygen deficit** of the mixture of stream water and wastewater

$$DO_0 = DO_s - DO$$

DO – actual dissolved oxygen at a given location downstream

$$DO\big(V(waste) + V(river)\big) = DO(waste)V(waste) + DO(river)V(river)$$

$$DO\times (V_w + V_r) = DO_w \times V_w + DO_r \times V_w$$

V = Q t V– volume; t – time; Q – volumetric flow rate

mg/L

$$\mathrm{DO}(\mathrm{Q_w} + \mathrm{Q_r}) = \mathrm{DO_w Q_w} + \mathrm{DO_r Q_r}$$

$$DO_0 = DO_S - \frac{Q_w DO_w + Q_r DO_r}{Q_w + Q_r}$$

Example 5.7 Initial oxygen deficit

The wastewater in previous example has a DO concentration of 2.0 mg/L and a discharge rate of 1.10 m^3/s. The river that is receiving this waste has DO equal to 8.3 mg/L, a flow rate of 8.70 m^3/s, and a temperature of 20 °C. Assuming complete and instantaneous mixing, estimate the initial DO deficit of the mixture of wastewater and river water just downstream from the discharge point.

Solution:

The initial amount of DO in the mixture of wastewater and river water would be

$$\mathrm{DO} = \frac{Q_w \mathrm{DO_w} + Q_r \mathrm{DO_r}}{Q_w + Q_r} = \frac{1.10 \times 2.0 + 8.70 \times 8.3}{1.10 + 8.70} = 7.6\ \mathrm{mg/L}$$

The saturated value of dissolved oxygen DO_s at 20 °C given in Table 5.10 as 9.09 mg/L, so the initial DO deficit would be:

$$D_0 = \mathrm{DO_s} - \mathrm{DO} = 9.09 - 7.6 = \mathbf{1.5\ mg/L}$$

Dissolved oxygen

You can calculate DOs based on this page.

TABLE 5.10

Solubility of Oxygen in Water (mg/L) at 1 atm Pressure

Temperature (°C)	Chloride Concentration in Water (mg/L)			
	0	5,000	10,000	15,000
0	14.62	13.73	12.89	12.10
5	12.77	12.02	11.32	10.66
10	11.29	10.66	10.06	9.49
15	10.08	9.54	9.03	8.54
20	9.09	8.62	8.17	7.75
25	8.26	7.85	7.46	7.08
30	7.56	7.19	6.85	6.51

Source: Thomann and Mueller, 1987.

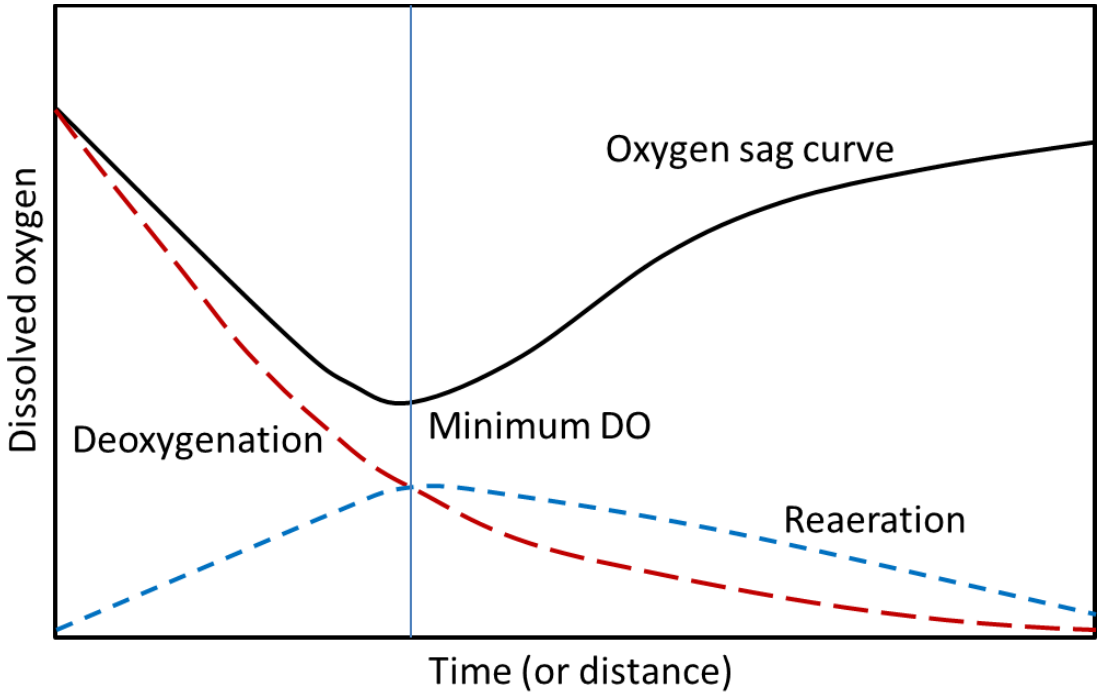
$DO_s = K_H P_{O2}$

Textbook Table 5.10 (page 213)

Oxygen sag curve

Rate of increase of deficit = rate of deoxygenation – rate of reaeration

$$DO = DO_s - \left[\frac{k_d L_0}{k_r - k_d}\left(e^{-k_d t} - e^{-k_r t}\right) + D_0 e^{-k_r t}\right]$$



$$t_c = \frac{1}{k_r - k_d} ln\left[\frac{k_r}{k_d}\left(1 - \frac{D_0(k_r - k_d)}{k_d L_0}\right)\right]$$

t – elapsed time between discharge point and distance x downstream.

t_c -- maximum deficit can then be found by substituting the value obtained for the critical time.

Rate of increase of deficit = Rate of deoxygenation – Rate of reaeration

Rate of deoxygenation = $k_d\, L_0 e^{-kdt}$

Rate of reoxygenation = $k_r D$ $\qquad D = DO_s - DO$

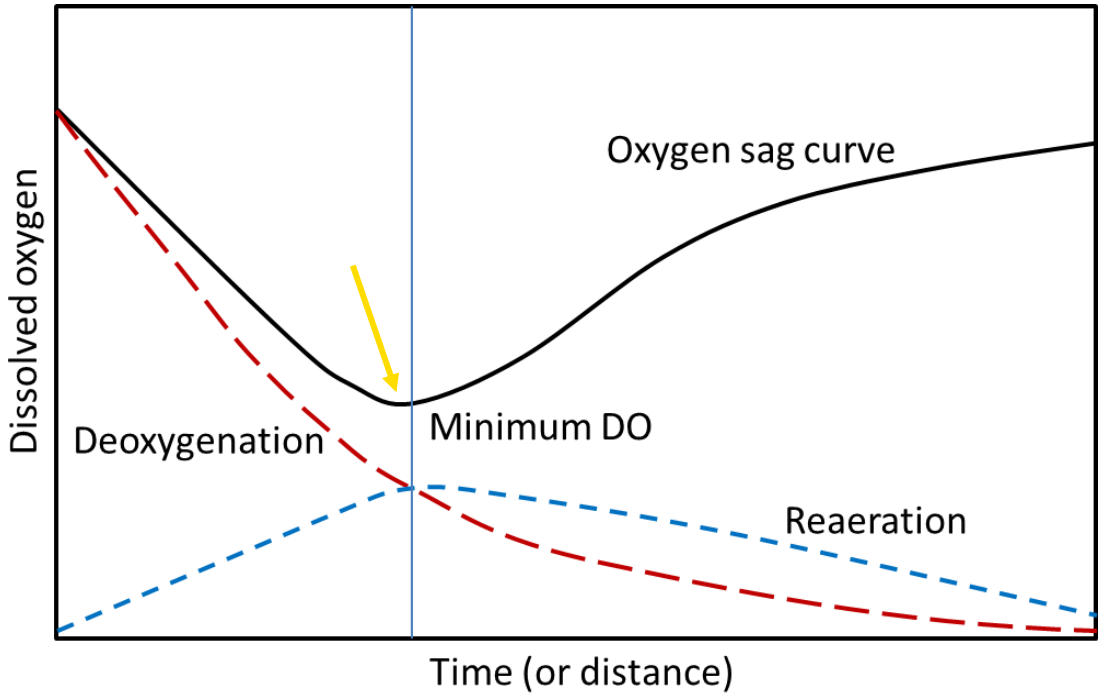
$$\frac{dD}{dt} = \mathrm{k_d}\, L_0 e^{-kdt} - k_r(DO_s - DO)$$

$$D = \frac{k_d L_0}{k_r - k_d}\left(e^{-k_d t} - e^{-k_r t}\right) + D_0 e^{-k_r t}$$

$$DO = DO_s - \left[\frac{k_d L_0}{k_r - k_d}\left(e^{-k_d t} - e^{-k_r t}\right) + D_0 e^{-k_r t}\right]$$

Setting the derivative of the oxygen deficit D = 0, then – the maximum deficit can then be found by substituting the value obtained for the critical time

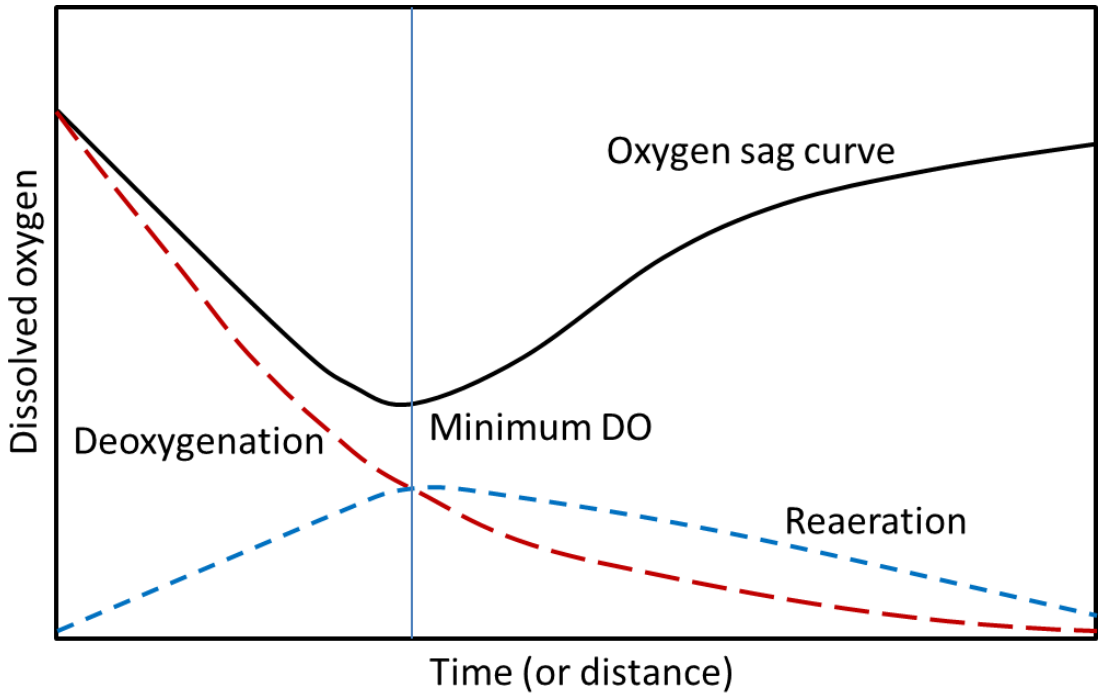
$$\frac{dD}{dt} = \mathrm{k_d}\, L_0 e^{-kdt} - k_r(DO_s - DO) = 0$$

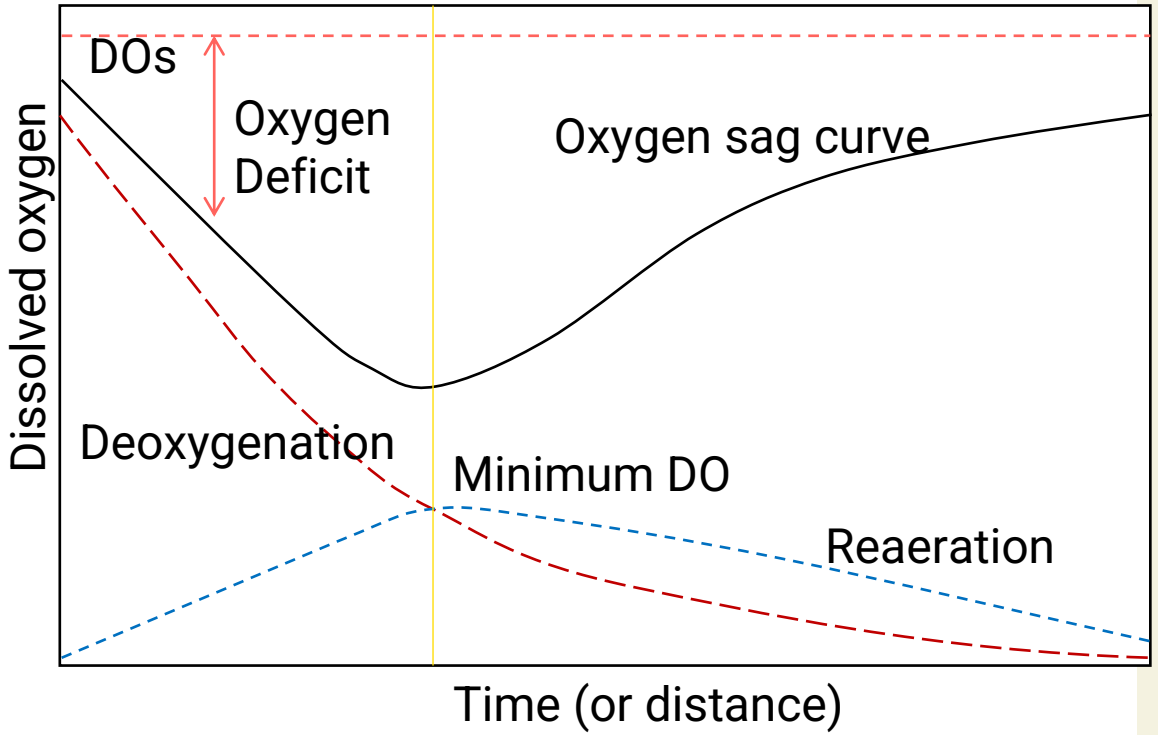


$$\mathrm{k_d}\, L_0 e^{-kdt} = k_r(DO_s - DO)$$

$$t_c = \frac{1}{k_r - k_d} ln \left[\frac{k_r}{k_d}\left(1 - \frac{D_0(k_r - k_d)}{k_d L_0}\right)\right]$$

Oxygen sag curve





Example 5.8 Streeter-Phelps oxygen sag curve

Just below the point where a continuous discharge of pollution mixes with a river, the BOD is 10.9 mg/L, and DO is 7.6 mg/L. The river and the wastewater mixture has a temperature of 20 °C, a deoxygenation constant k_d of 0.20/day, an average flow speed of 0.30 m/s, and an average depth of 3.0 m.

a. Find the time and distance downstream at which the oxygen deficit is at a maximum.
b. Find the minimum value of DO.

Solutions:

From Table 5.10, DO_s at 20 is 9.1 mg/L, so the initial DO deficit is:

$$D_0 = \mathrm{DO_s} - \mathrm{DO} = 9.09 - 7.6 = 1.5\ \mathrm{mg/L}$$

To estimate the reaeration constant, we can use the equation of:

$$k_r = \frac{3.9u^{1/2}}{H^{3/2}} = \frac{3.9\times 0.30^{1/2}}{3.0^{3/2}} = 0.41/\mathrm{day}$$

a. The time at which the deficit is a maximum would be:

$$t_c = \frac{1}{k_r - k_d}\ln\left\{\frac{k_r}{k_d}\left[1 - \frac{D_0(k_r - k_d)}{k_d L_0}\right]\right\} = \frac{1}{0.41 - 0.20}\ln\left\{\frac{0.41}{0.20}\left[1 - \frac{1.5\times(0.41 - 0.20)}{0.20\times 10.9}\right]\right\}$$

$$= \mathbf{2.67\ days}$$

So the critical distance downstream would be

$$x_c = ut_c = 0.30\times 2.67\times 86400 = \mathbf{69.3\ km}$$

Example

Just below the point where a continuous discharge of pollution mixes with a river, the BOD is 10.9 mg/L, and DO is 7.6 mg/L. The river and the wastewater mixture has a temperature of 20 °C, a deoxygenation constant k_d of 0.20/day, an average flow speed of 0.30 m/s, and an average depth of 3.0 m.

a. Find the time and distance downstream at which the oxygen deficit is at a maximum.
b. Find the minimum value of DO.

Solutions:

b. The maximum deficit would be:

$$D = \frac{k_d L_0}{k_r - k_d}\left(e^{-k_d t} - e^{-k_r t}\right) + D_0 e^{-k_r t} = \frac{0.20 \times 1.09}{0.41 - 0.20}\left(e^{-0.20 \times 2.67} - e^{-0.41 \times 2.67}\right) + 1.5 e^{-0.41 \times 2.67}$$

$$= \mathbf{3.1\ mg/L}$$

So the minimum value of DO will be the saturation value minus this maximum deficit:

$$\mathrm{DO_{min}} = \mathrm{DO_s} - D = 9.1 - 3.1 = \mathbf{6.0\ mg/L}$$

5.5 Water Phenomenon & Explaination

- **Eutrophication**

  ----Eutrophication, nutrients (N & P), limiting nutrient

- **Thermal stratification**

Water quality in lakes and reservoirs



aol.research.pdx.edu



http://en.wikipedia.org/wiki/Eutrophication

seaweed ⟹

cyanobacteria ⟹ chlorophyll

An oligotrophic lake:
Low in nutrients; Low production level; Less turbid; Higher DO; More diversity.

Water quality in lakes and reservoirs

Eutrophication – "The enrichment of bodies of fresh water by inorganic plant nutrients (e.g. nitrate, phosphate). It may occur **naturally** but can also be the result of **human activity** (cultural eutrophication from fertilizer runoff and sewage discharge) and is particularly evident in slow-moving rivers and shallow lakes … Increased sediment deposition can eventually raise the level of the lake or river bed, allowing land plants to colonize the edges, and eventually converting the area to dry land." - Lawrence and Jackson, 1998

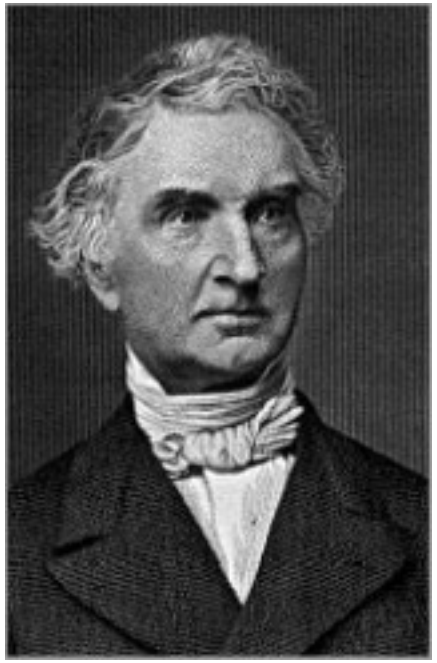
Lawrence, E., Jackson, A.R.W., and Jackson, J.M., 1998, Eutrophication, *in* Longman Dictionary of Environmental Science: London, England, Addison Wesley Longman Limited, p. 144-145.

Eutrophication – over supply of nutrients (usually N and P) to a lake, causing the overgrowth of aquatic life

Limiting nutrient

Law of the minimum nutrient

"growth of a plant is dependent on the amount of foodstuff that is presented to it in minimum quantity" – Justus Liebig



Father of Fertilizer industry

Essential nutrients for algal growth

- C, N, P, S, Ca, Mg, K, Na, Fe, Mn, Zn, Cu, B, …
- Limiting nutrients: typically N or P

Limiting nutrient

A representative “chemical formula” for microalgae
$C_{106}H_{263}O_{110}N_{16}P$

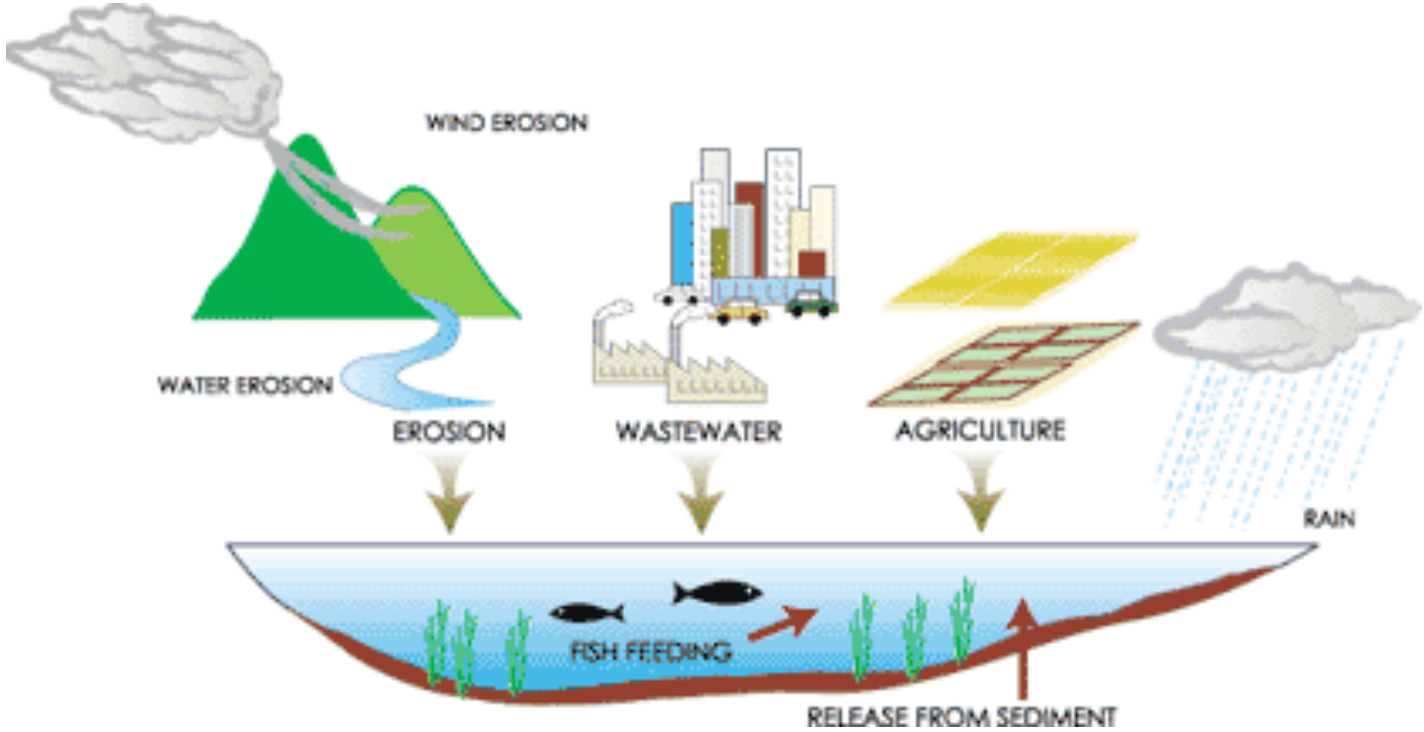
N to P weight ratio: N/P = (14 x 16)/31 = 7.2

The above ratio varies depend on algal species

Rule of thumb

- N/P > 10 → P limiting
- N/P < 10 → N limiting

Limiting nutrient



http://www.unep.or.jp/ietc/publications/short_series/lakereservoirs-3/3.asp

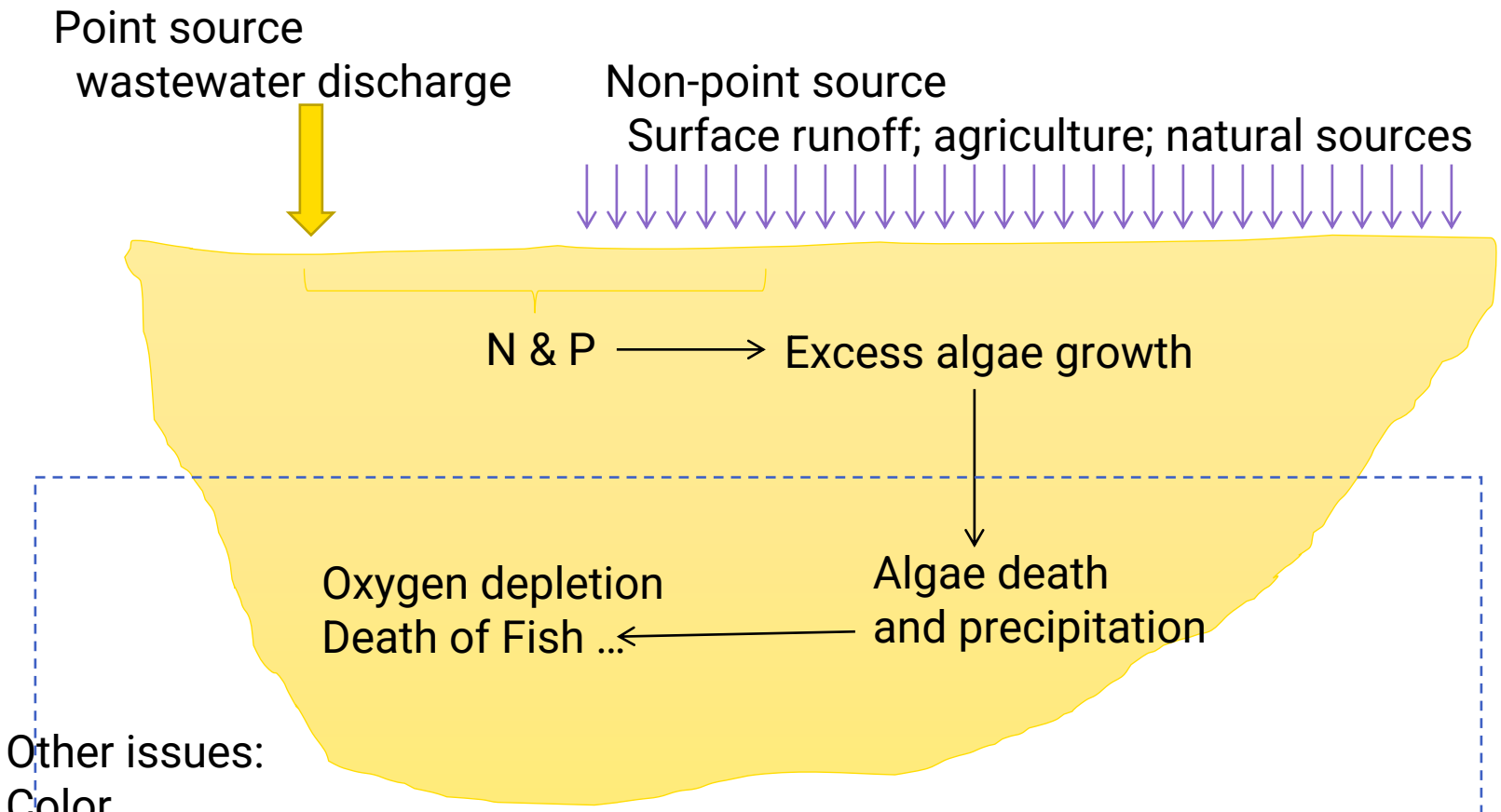
→ the quickest way to control eutrophication is to control nutrient

P: 0.010 mg/L is acceptable; 0.020 mg/L is excessive (Vollenweider, 1975)

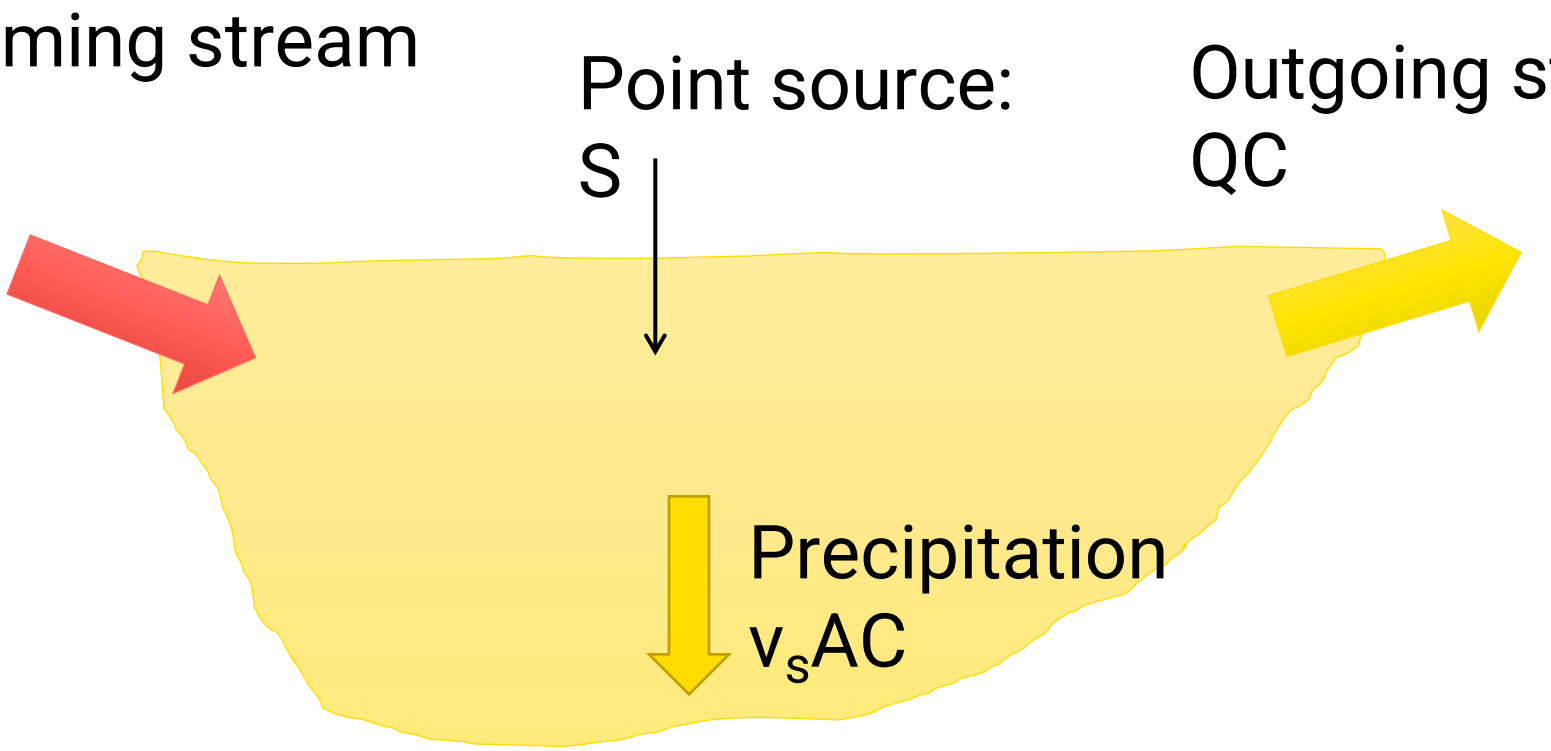
NASA Discovering dead zones in the oceans and lakes

http://www.youtube.com/watch?v=XEZpo9uLIc0

Limiting nutrient



Phosphorus model



$$QC_{in} + S = QC + v_sAC \quad \rightarrow \quad C = \frac{QC_{in} + S}{Q + v_sA}$$

Read Ex. 5.9

$$QC_{in} + S = QC + v_s AC \quad \Rightarrow \quad C = \frac{QC_{in} + S}{Q + v_s A}$$

S = the rate of addition of phosphorus from the point-source (g/s)

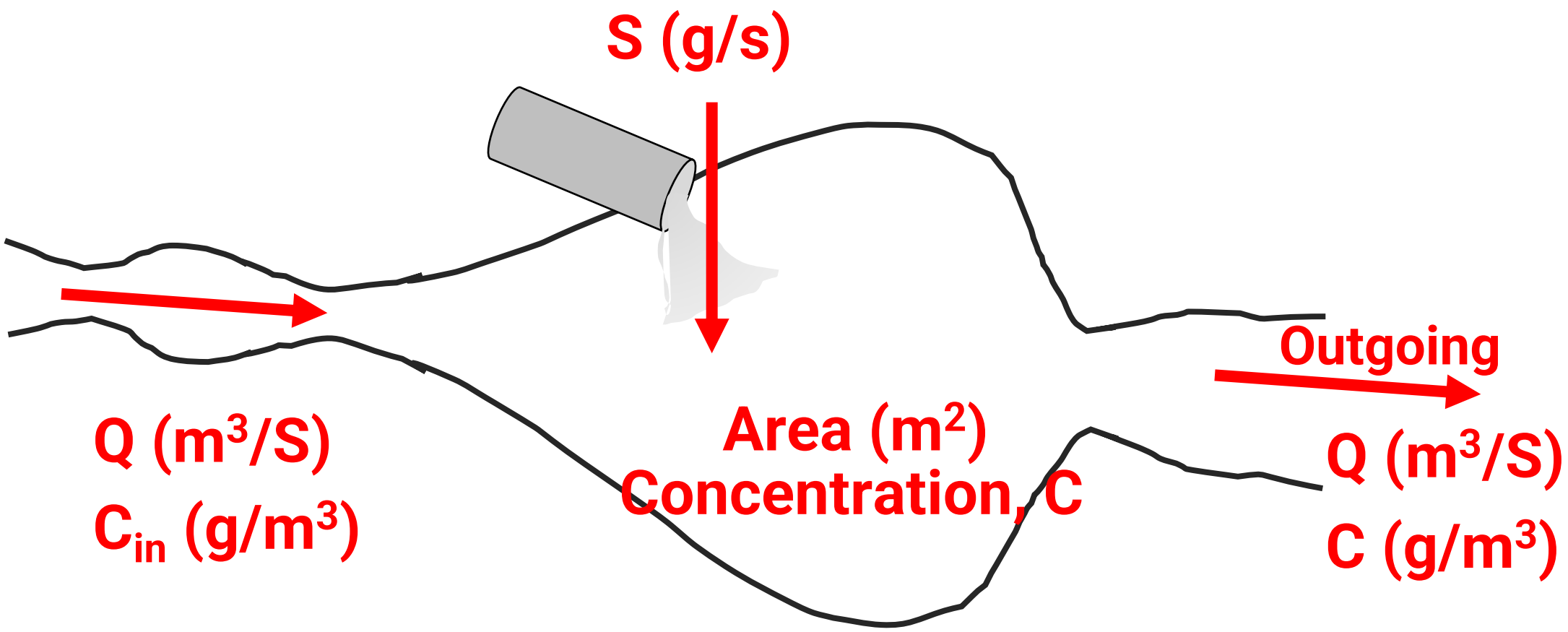
C = the concentration of phosphorus in the lake (g/m?)

Cin = the concentration of phosphorus in the incoming stream flow (g/m3)

Q = the stream inflow and outflow rate (m3/s)

Vs = the phosphorus settling rate (m/s)

A = the surface area of the lake (m2)


S (g/s)
Outgoing
Q (m^3/S)
C_{in} (g/m^3)
Area (m^2)
Concentration, C
Q (m^3/S)
C (g/m^3)

Examples

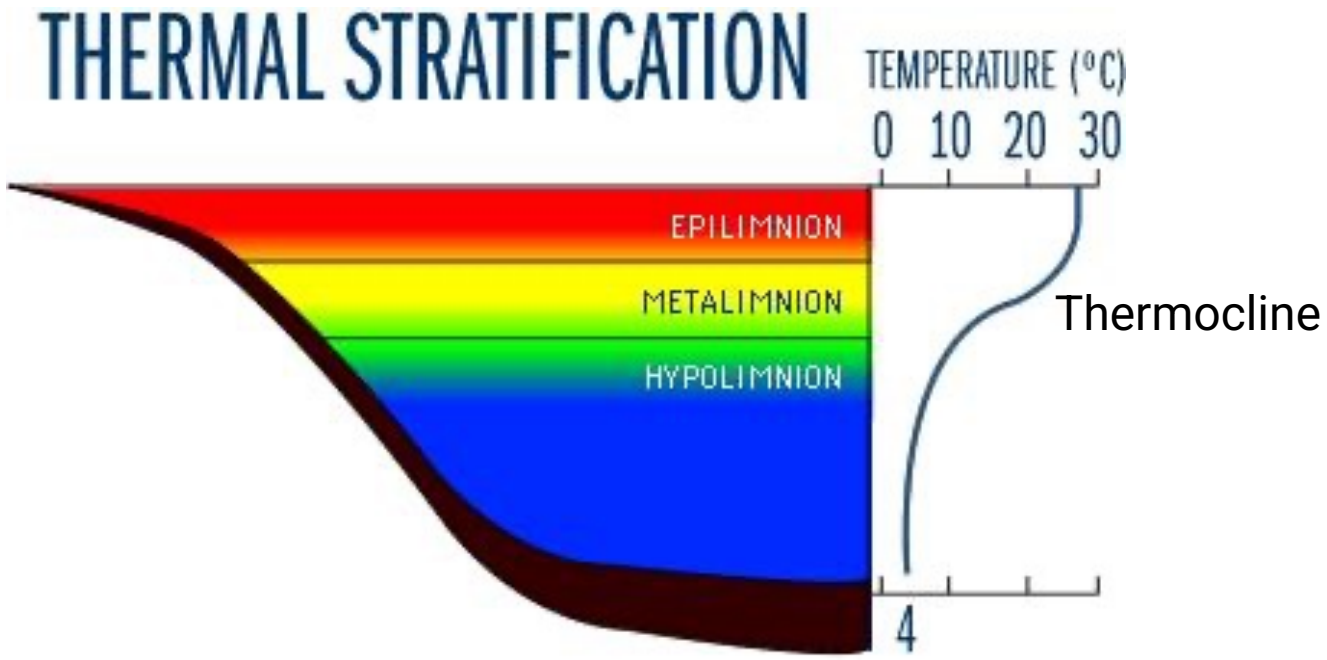
1. Eutrophication is an emerging global issue associated with increasing anthropogenic nutrient loading. Aquatic eutrophication facilitates blooms of cyanobacteria including cyanotoxin producing species, which profoundly impact aquatic ecosystems and human health.

1) Briefly discuss the eutrophication as well as its adverse effects in rivers and ocean, respectively **(< 300 words**).

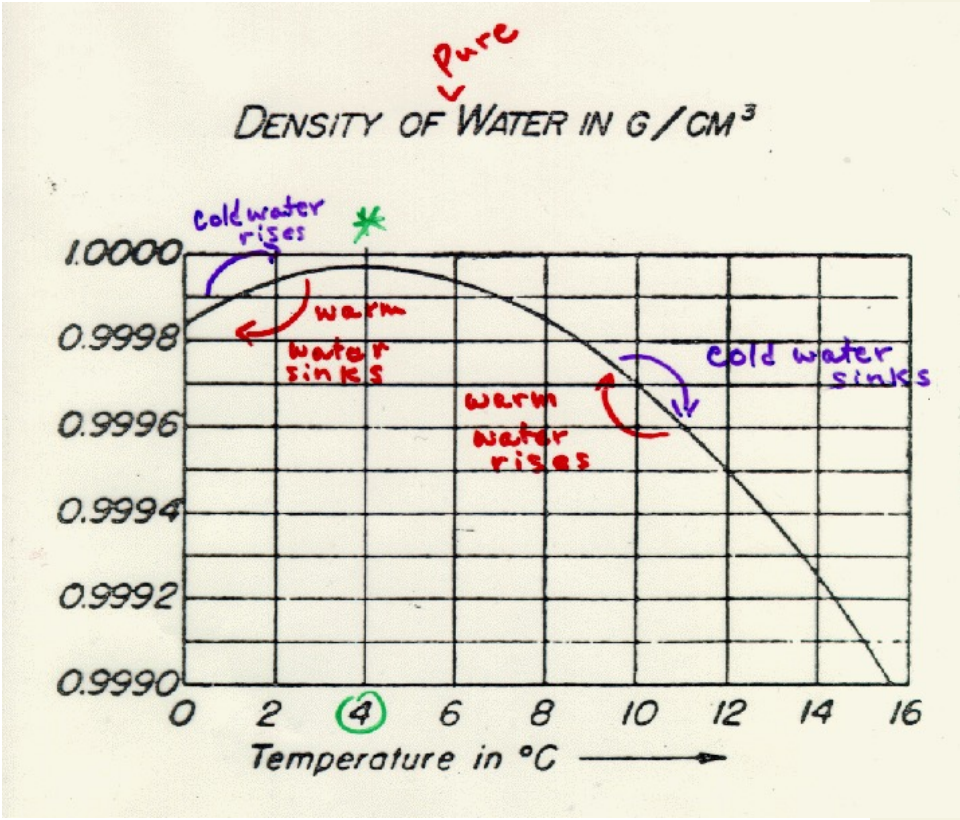
Reasons and effects

2) An aquatic eutrophication happened in a lake with 2820 km^2, which mainly due to the excessive discharge of nutrients, please design an appropriate treatment process to reduce the discharge of nutrients, thereby mitigating water eutrophication (< **200 words including explanation of your design purposes**, **up to 1 Figure is allowed**), and preventive measures (< **100 words including explanation of your purposes**).

Thermal stratification



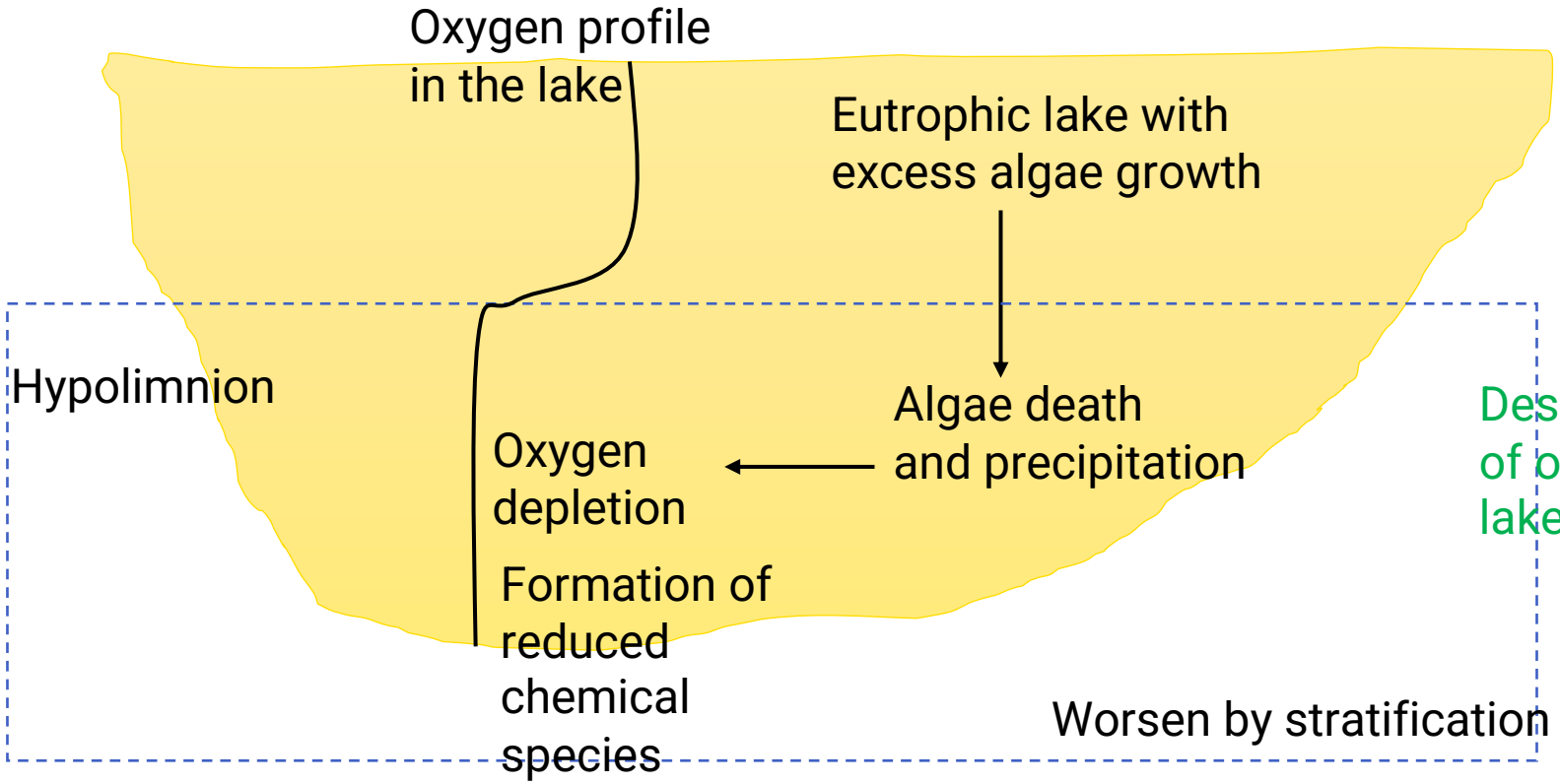
http://msue.anr.msu.edu/news/the_thermocline_a_summer_phenomenon_in_michigan_inland_lakes



The lower layer of water in a stratified lake, typically coder than the water above and relatively stagnant.

Thermal stratification in eutrophic lake

DO is high in the epilimnion due to reaeration and photosynthesis. The hypolimnion, on the other hand, is cut off from the oxygen-rich epilimnion by stratification with low DO.



Describe the distribution of organism in eutrophic lake

Dissolved oxygen is greatly affected by both eutrophication and thermal stratification.

Example from past examination

A factory generates 6.0×10^4 L/min wastewater discharged into a river with a flow of 8.0×10^5 m^3/d. At 20 °C, the wastewater has a DO concentration of 2.5 mg/L and an ultimate BOD of 70.0 mg/L, while the river water has a DO concentration of 7.5 mg/L and an ultimate BOD of 5.0 mg/L. The deoxygenation constant is 0.20 day^{-1} and the saturated value of DO is 9.1 mg/L at 20 °C.

(c) Assuming that the wastewater and river water are well-mixed at the discharge point, find the initial DO deficit of the mixture just downstream the discharge point.

(d) Assuming that the wastewater and river water are well-mixed at the discharge point, estimate the BOD_5 value of the mixture just downstream the discharge point at 25 °C.

2020

Readings and practice questions

Readings

- Essential reading: pp173-226

Practice questions

- Q5.5, 5.6, 5.12, 5.17, 5.35